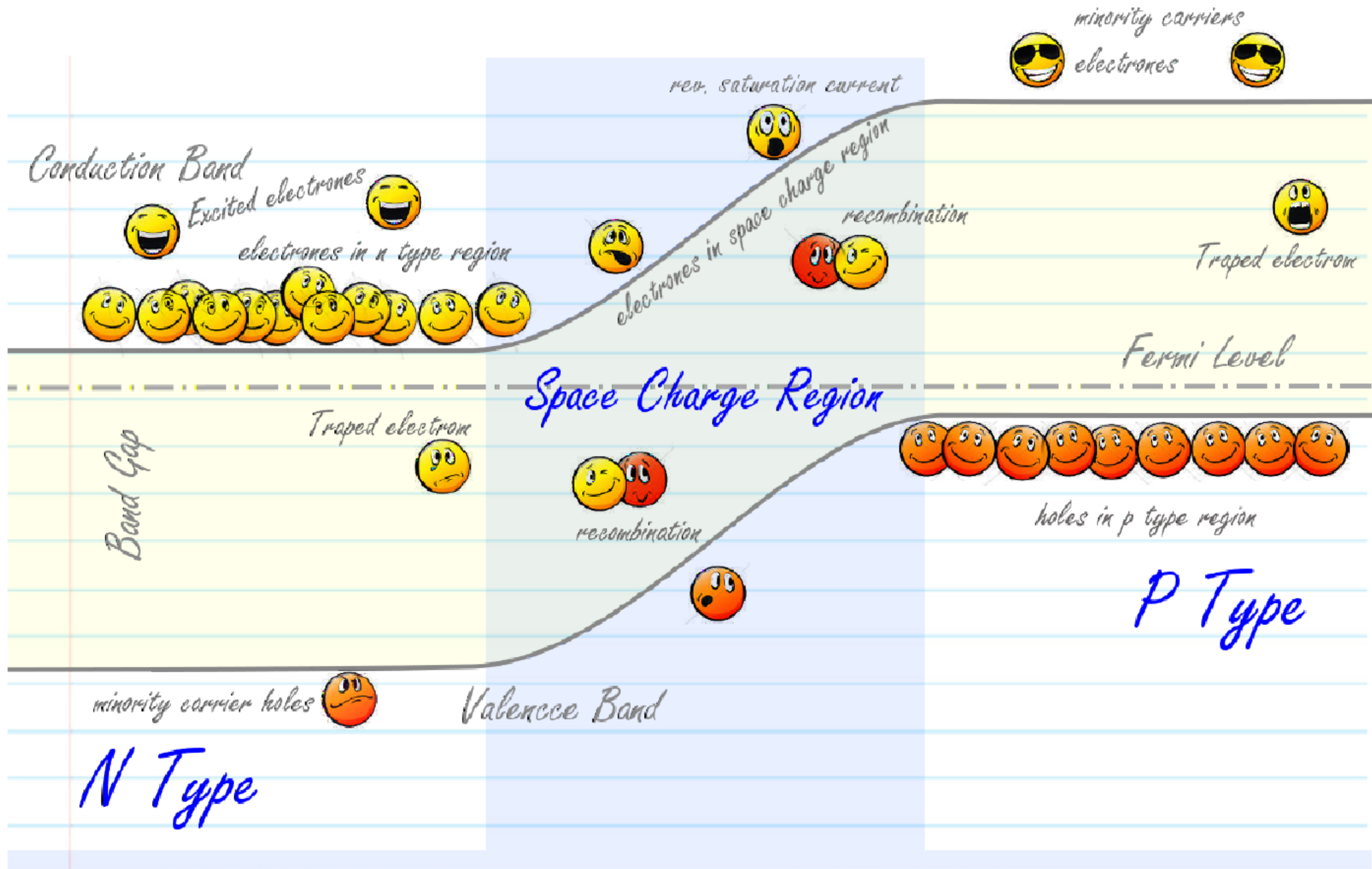


PN Junction



content

- abrupt pn-junction
- depletion region
- built-in potential
- biased junction

Where do I see PN diodes everyday ?

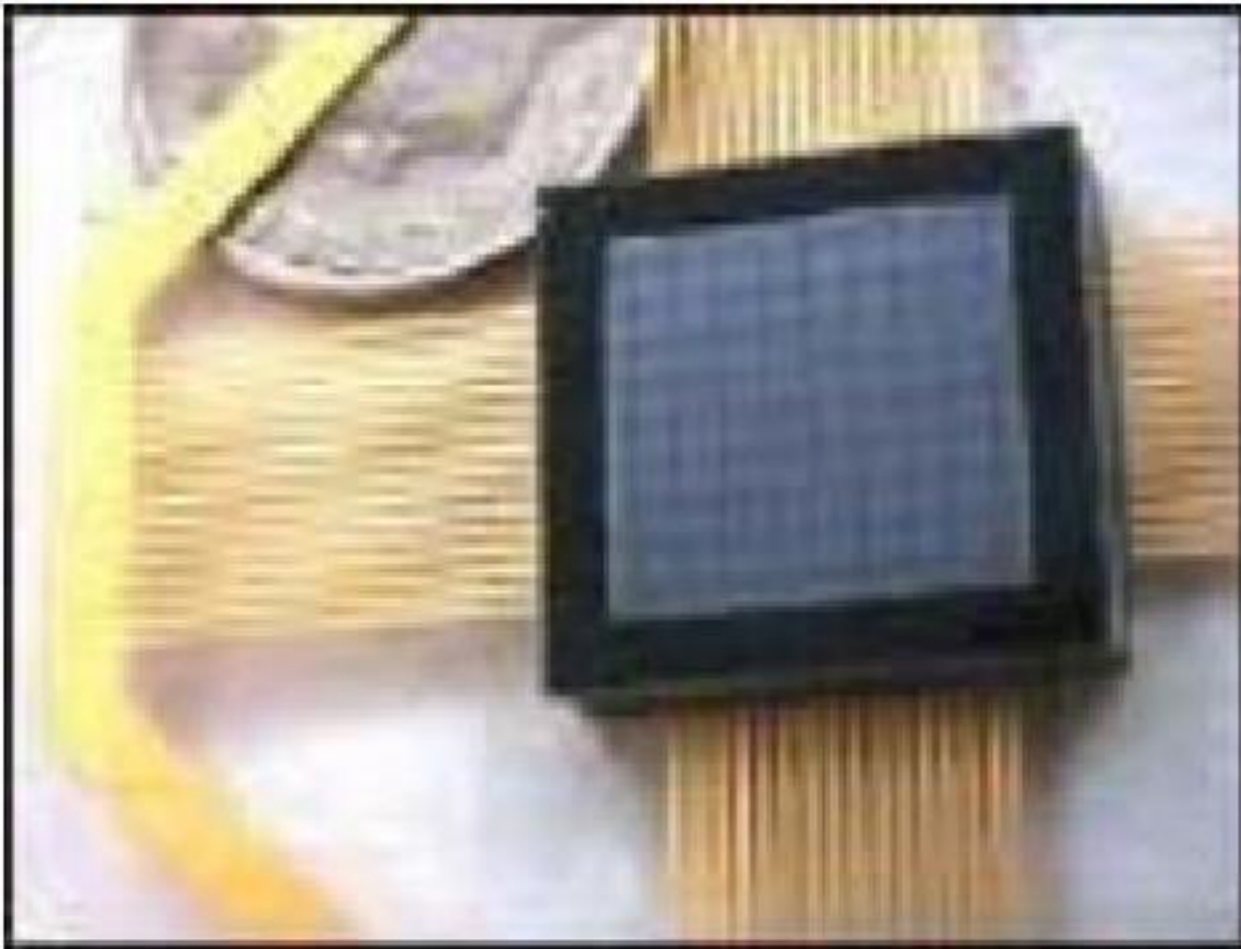
solar cells



laser



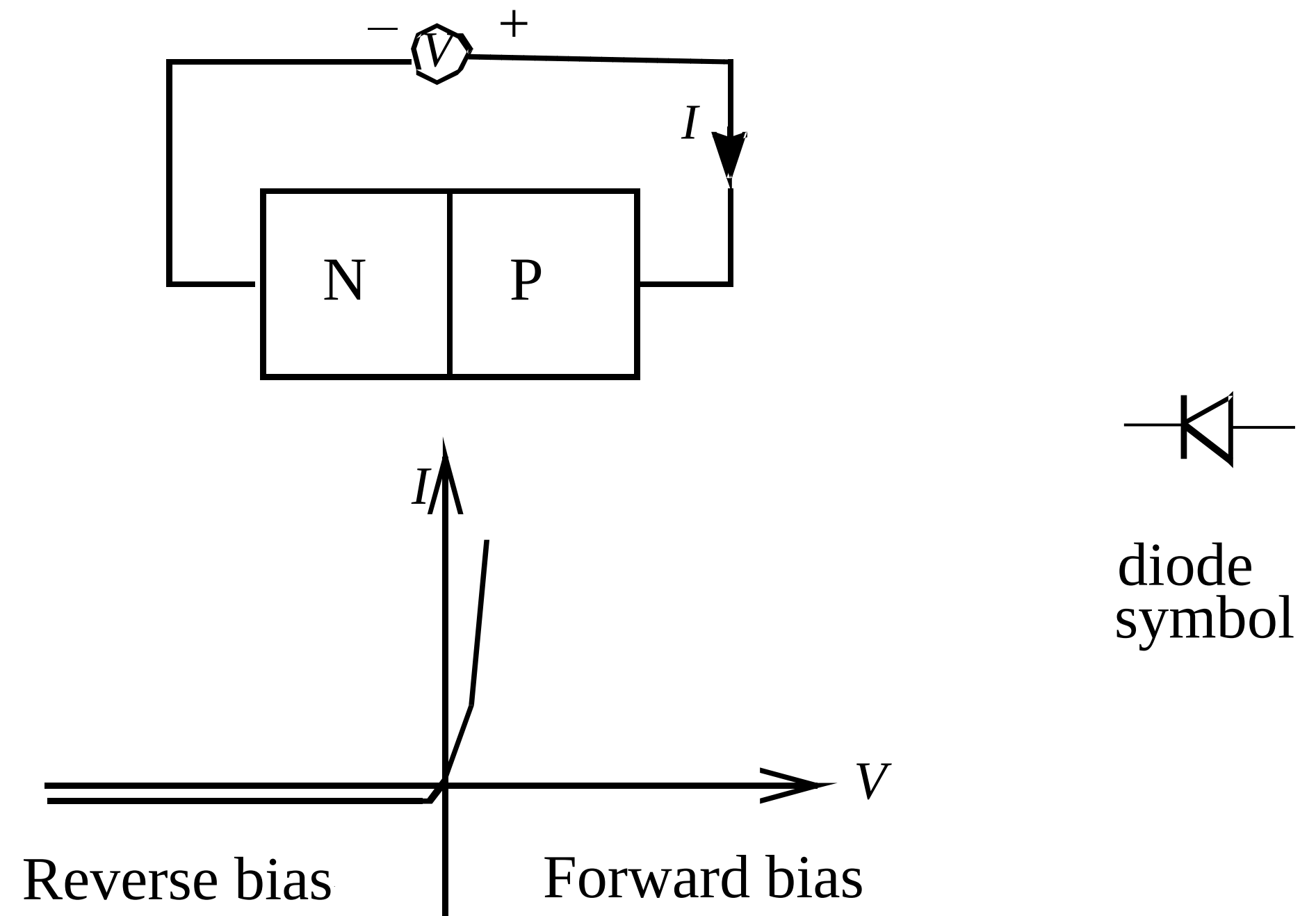
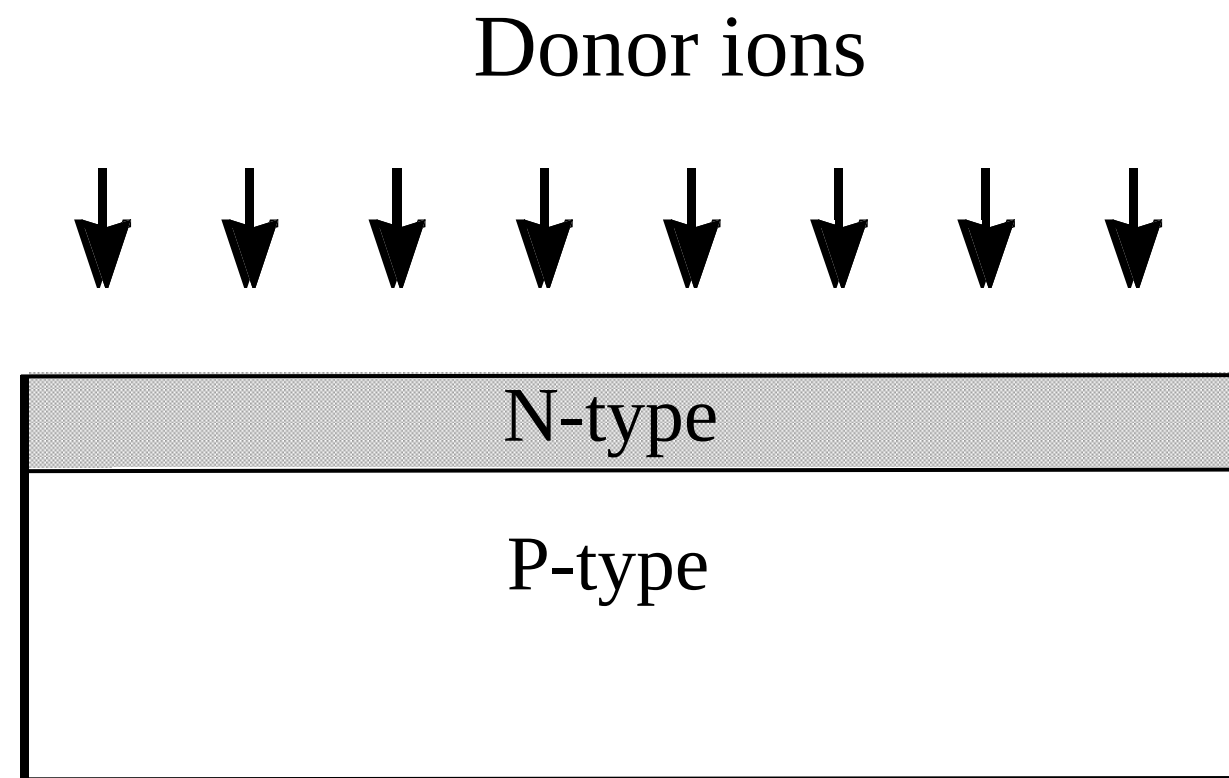
Avalanche Photodiode



OLED display



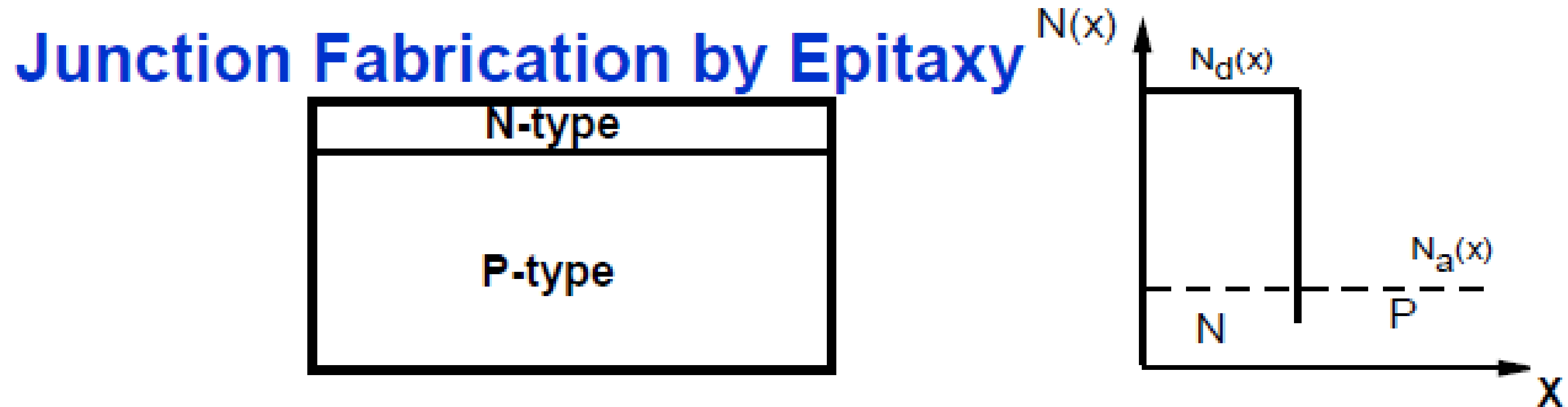
Building Blocks for semiconductor devices



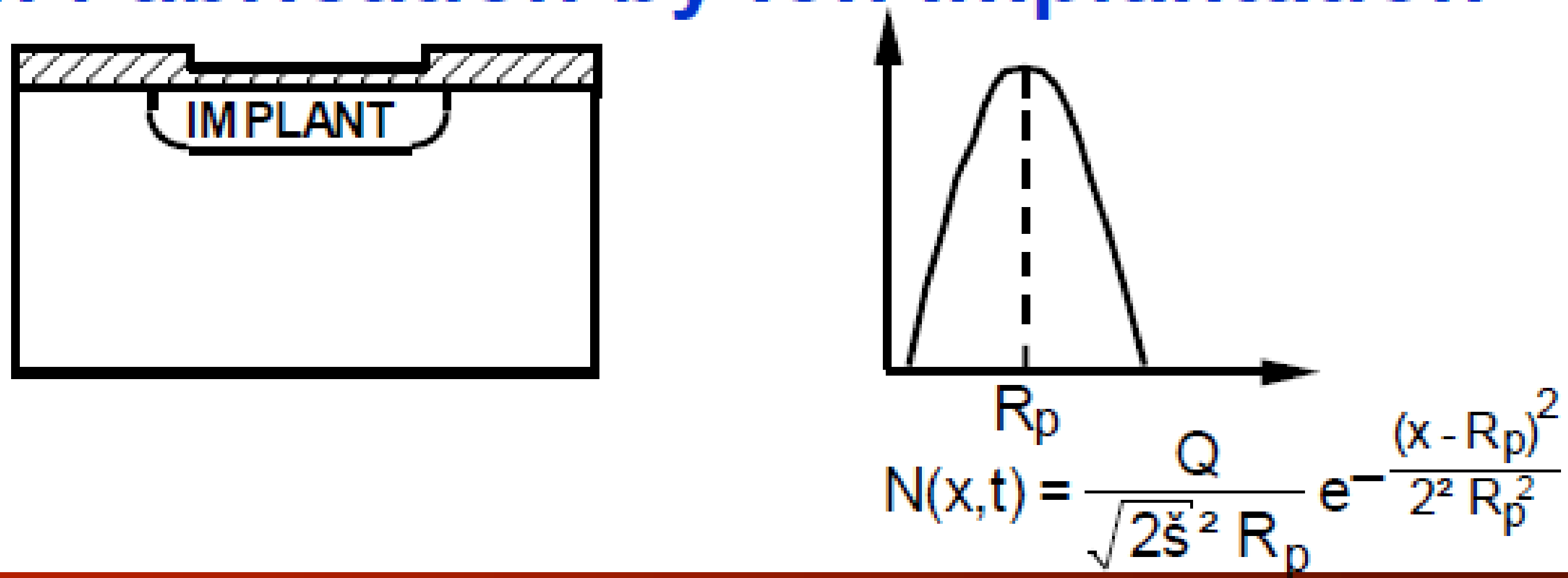
PN junction is present in perhaps every semiconductor device.

P-N Junction Formation

P-N junctions, in addition to being a useful semiconductor device, form the basis for almost all other semiconductor devices. Understanding their operation is basic to understanding most devices

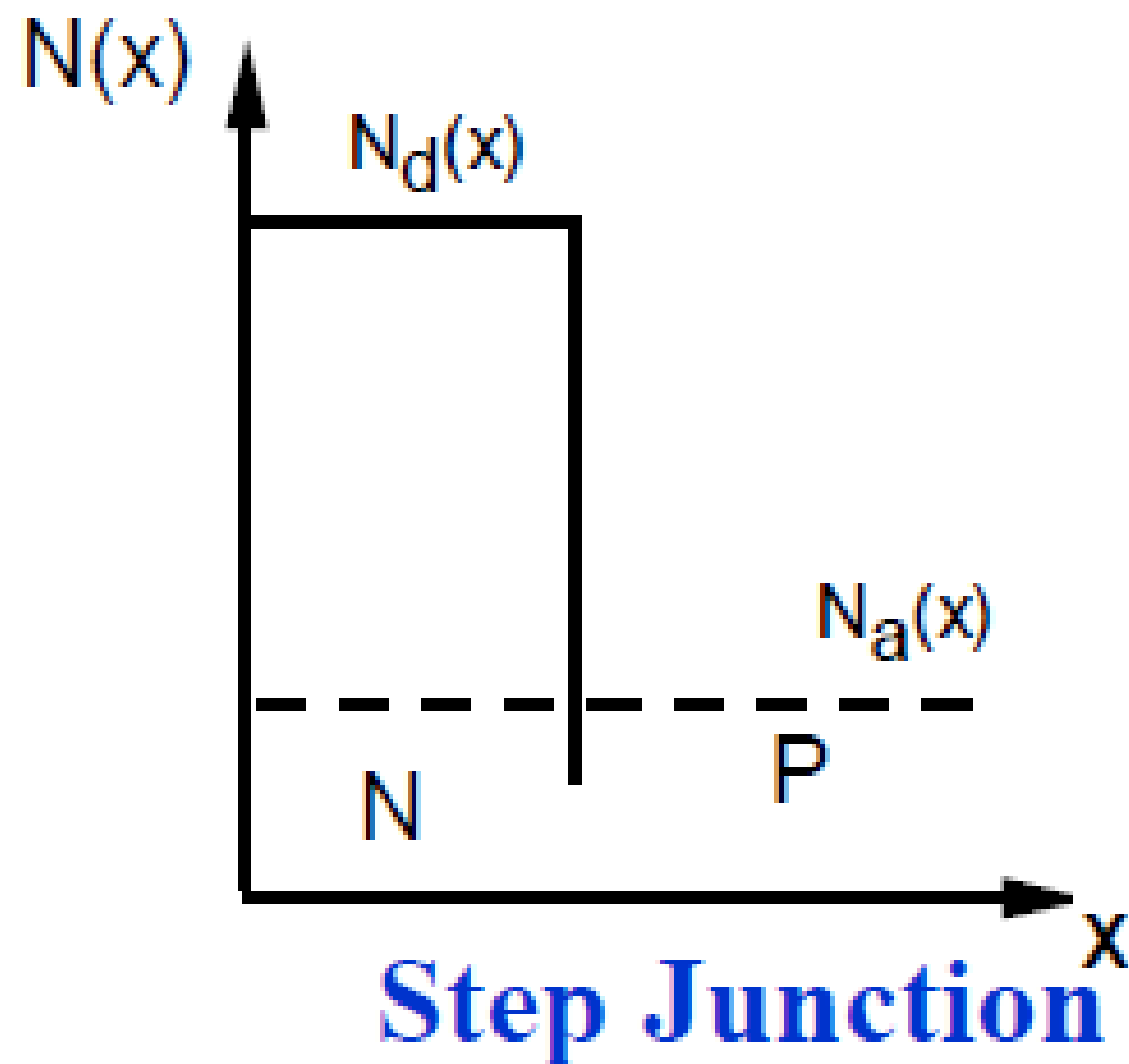


Junction Fabrication by Ion Implantation

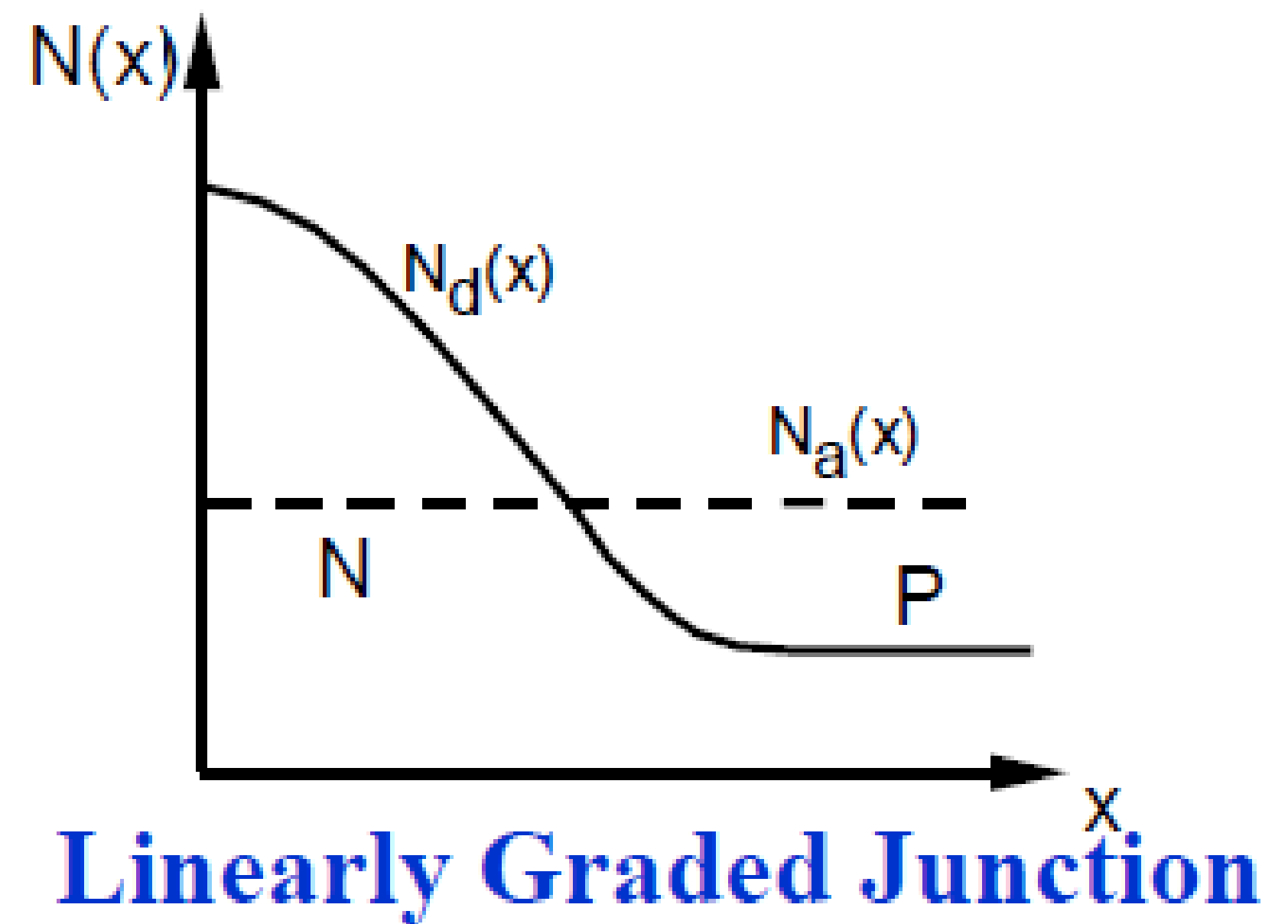


P-N Junction Profiles

The two profiles that are encountered most often in real devices are also easiest to analyze from a device physics point of view

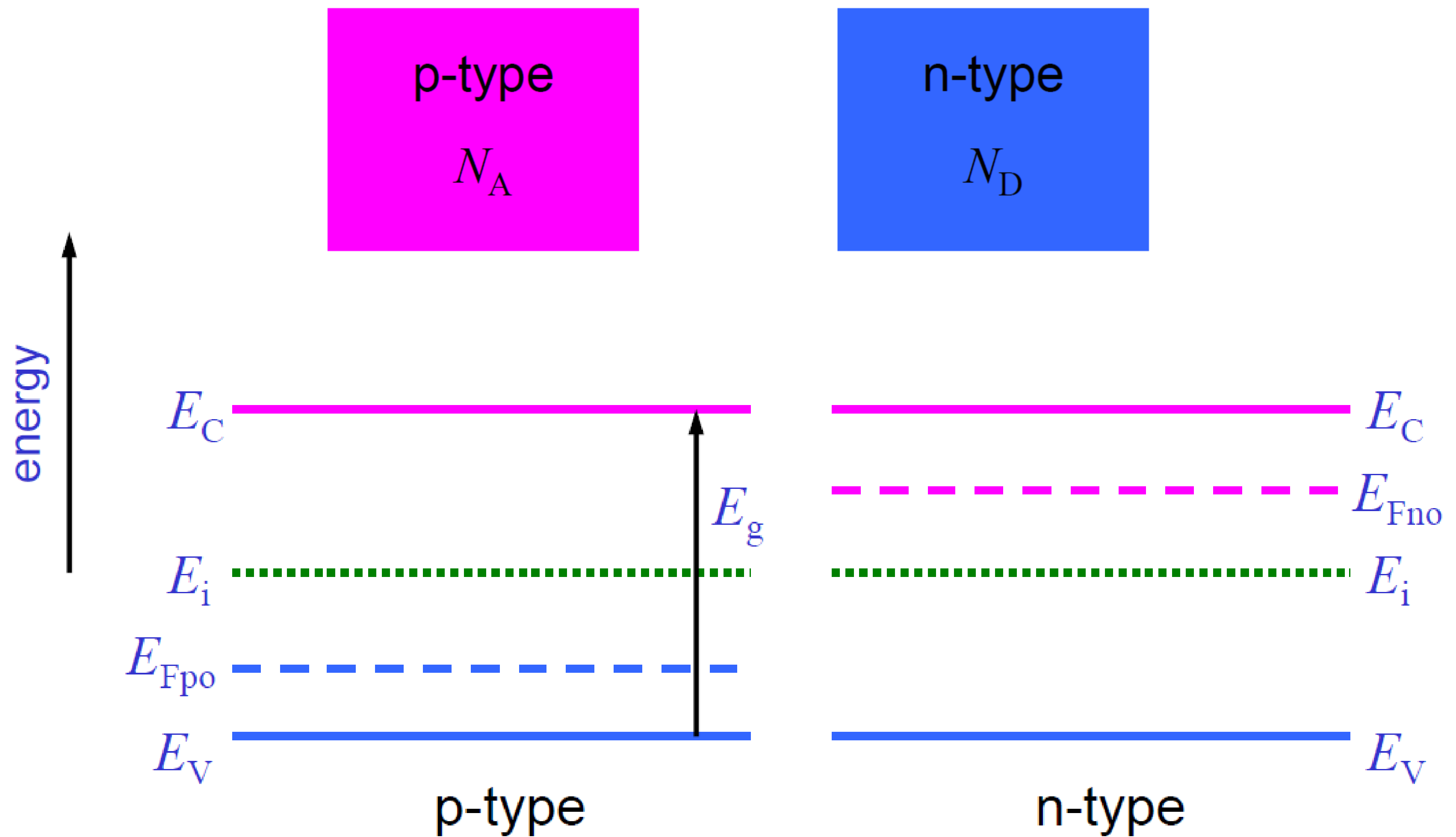


Good approximation for shallow, high concentration junctions ($x_j < 1\mu\text{m}$) and epitaxially grown junctions



Good approximation for deep junctions ($x_j > 3\mu\text{m}$)

Before connection



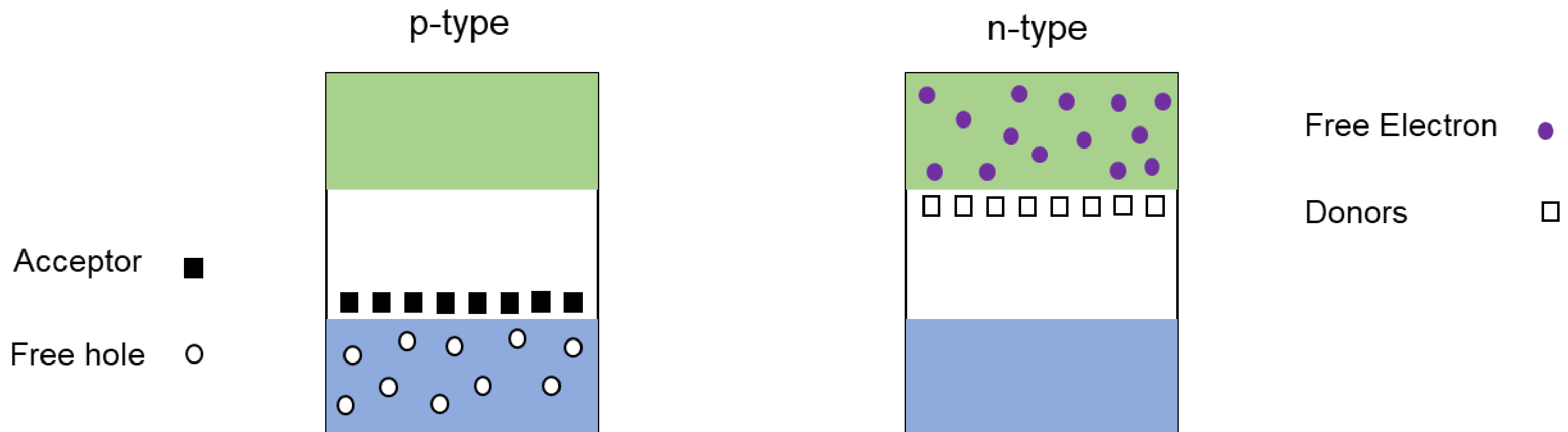
P-N Junction in Equilibrium

The equilibrium condition implies:

- No application of any external voltage
- No light shining (dark conditions)
- Uniform temperature through out the semiconductor
- No externally applied electrical or magnetic fields

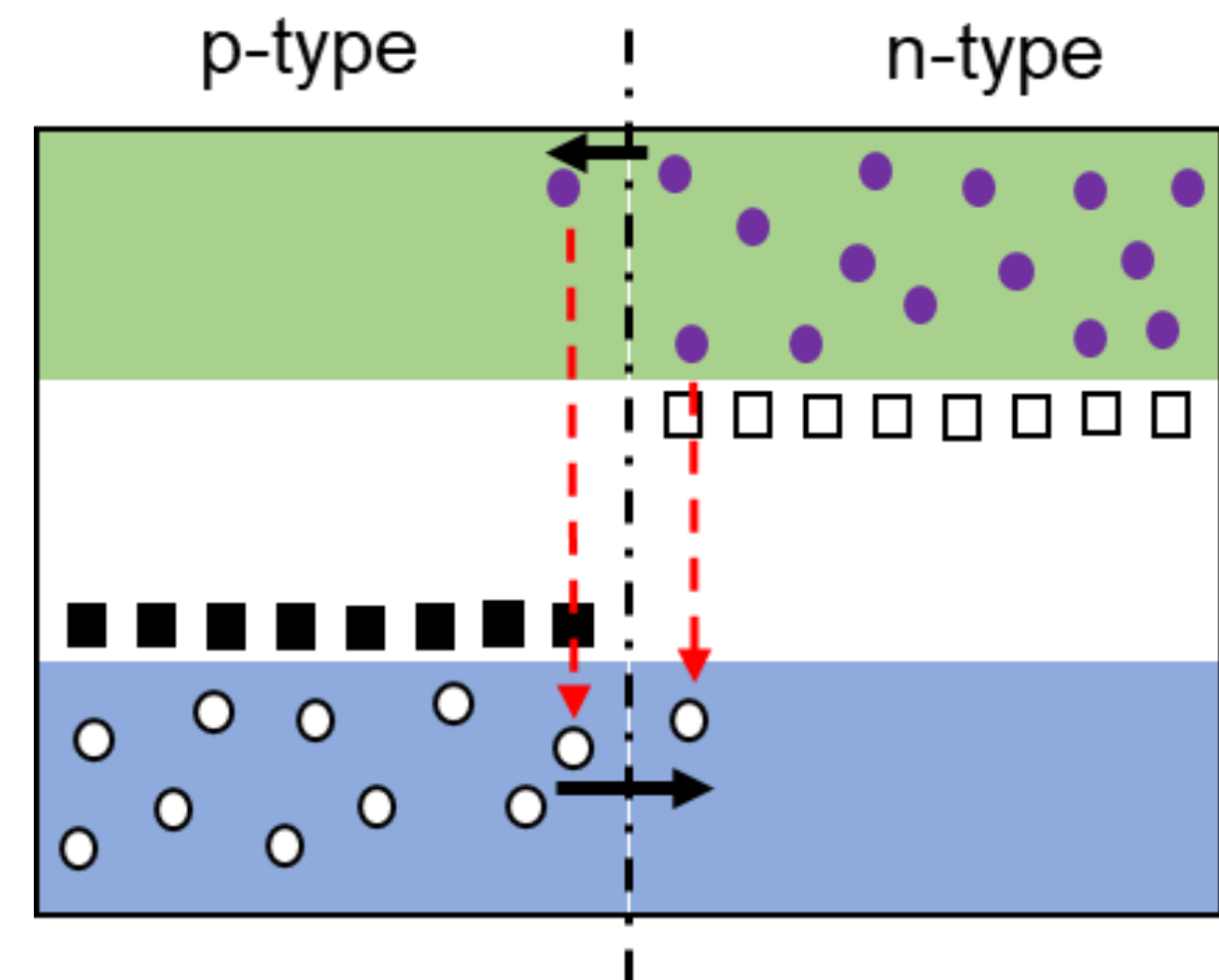
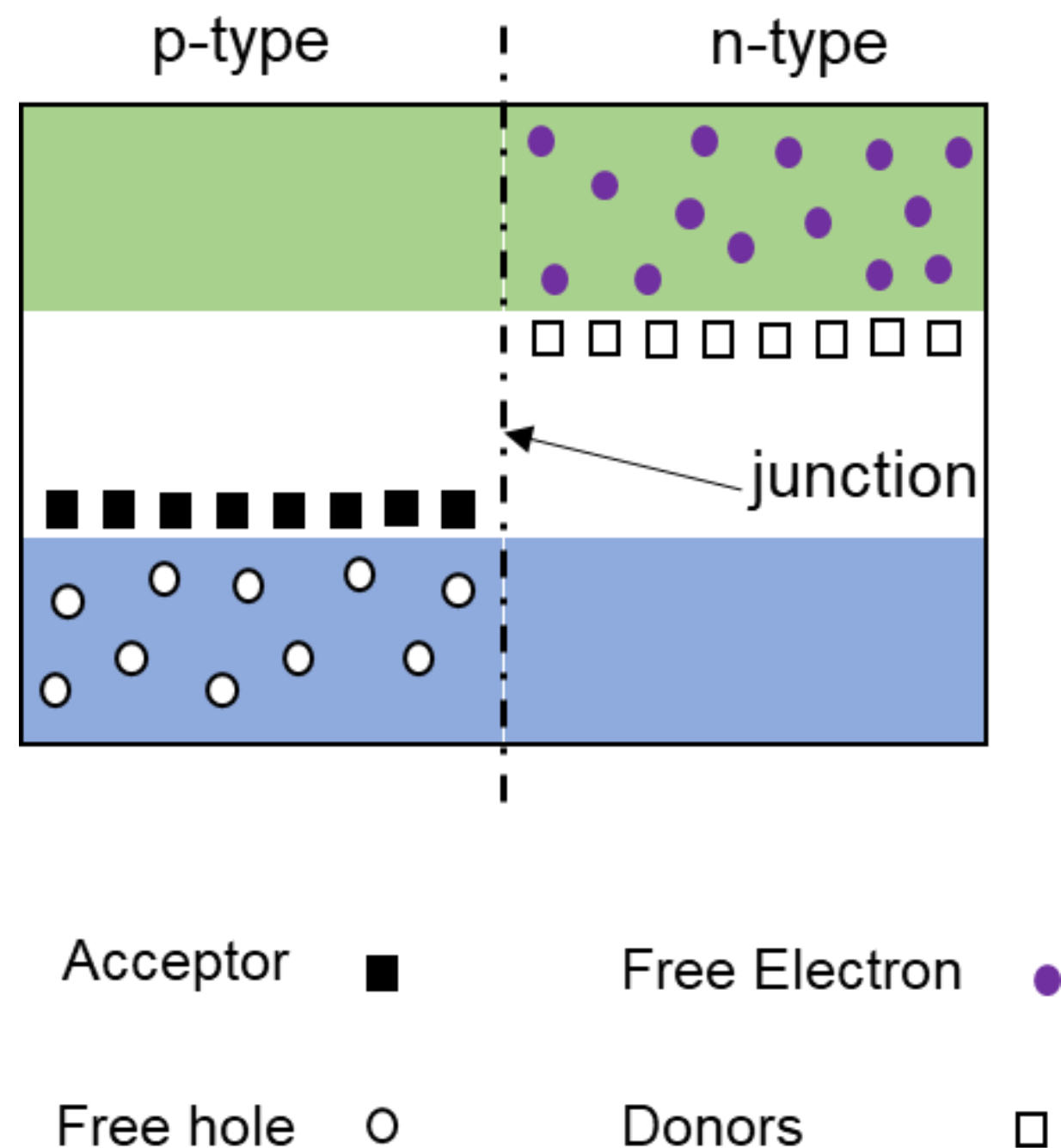
In short, the p-n junction is maintained at a constant temperature with NO external disturbance of any kind.

To understand how a p-n junction diode works, begin by imagining two separate bits of semiconductor, one n-type, the other p-type.

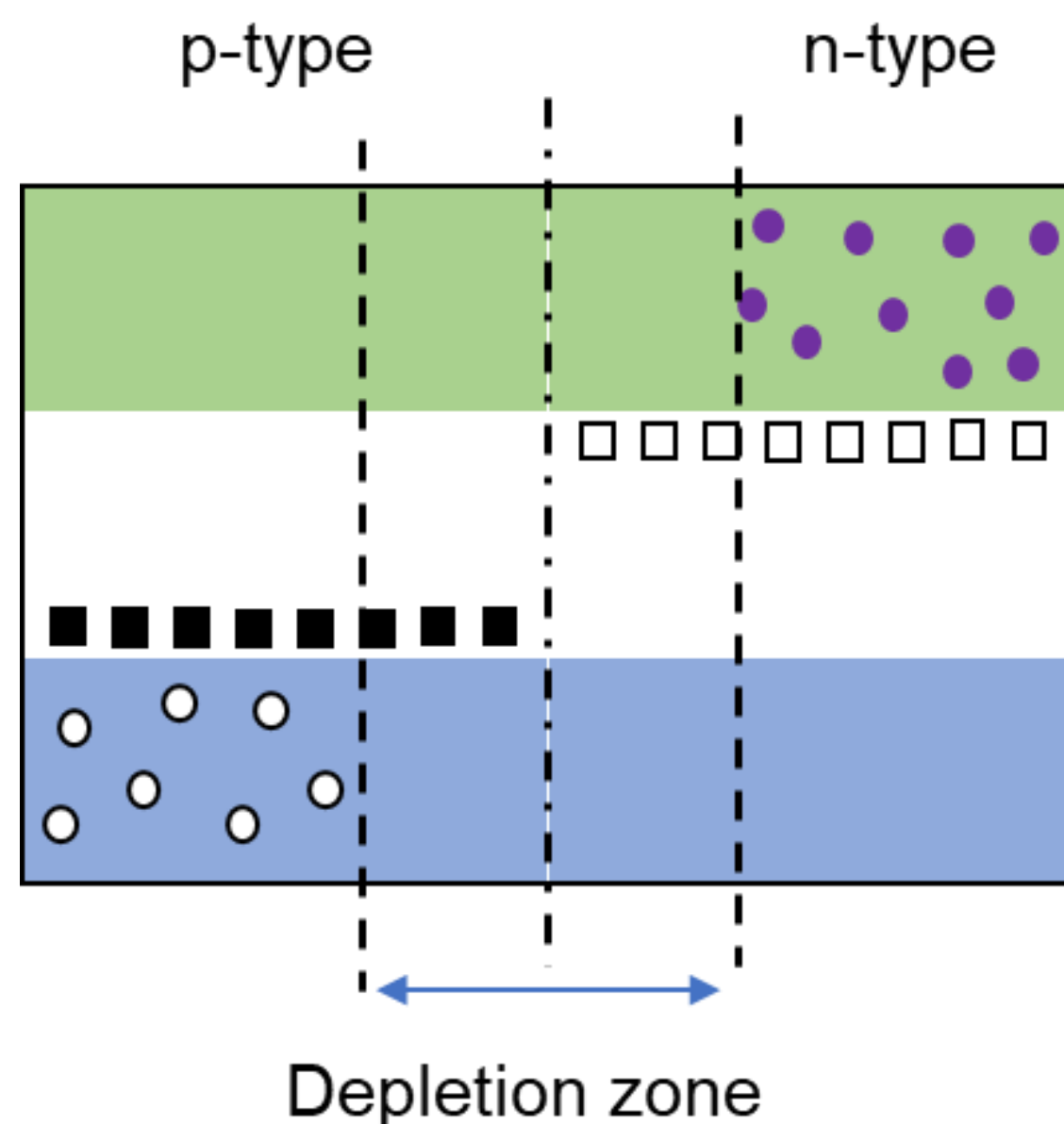


Bring them together and join them to make one piece of semiconductor which is doped differently on either side of the junction.

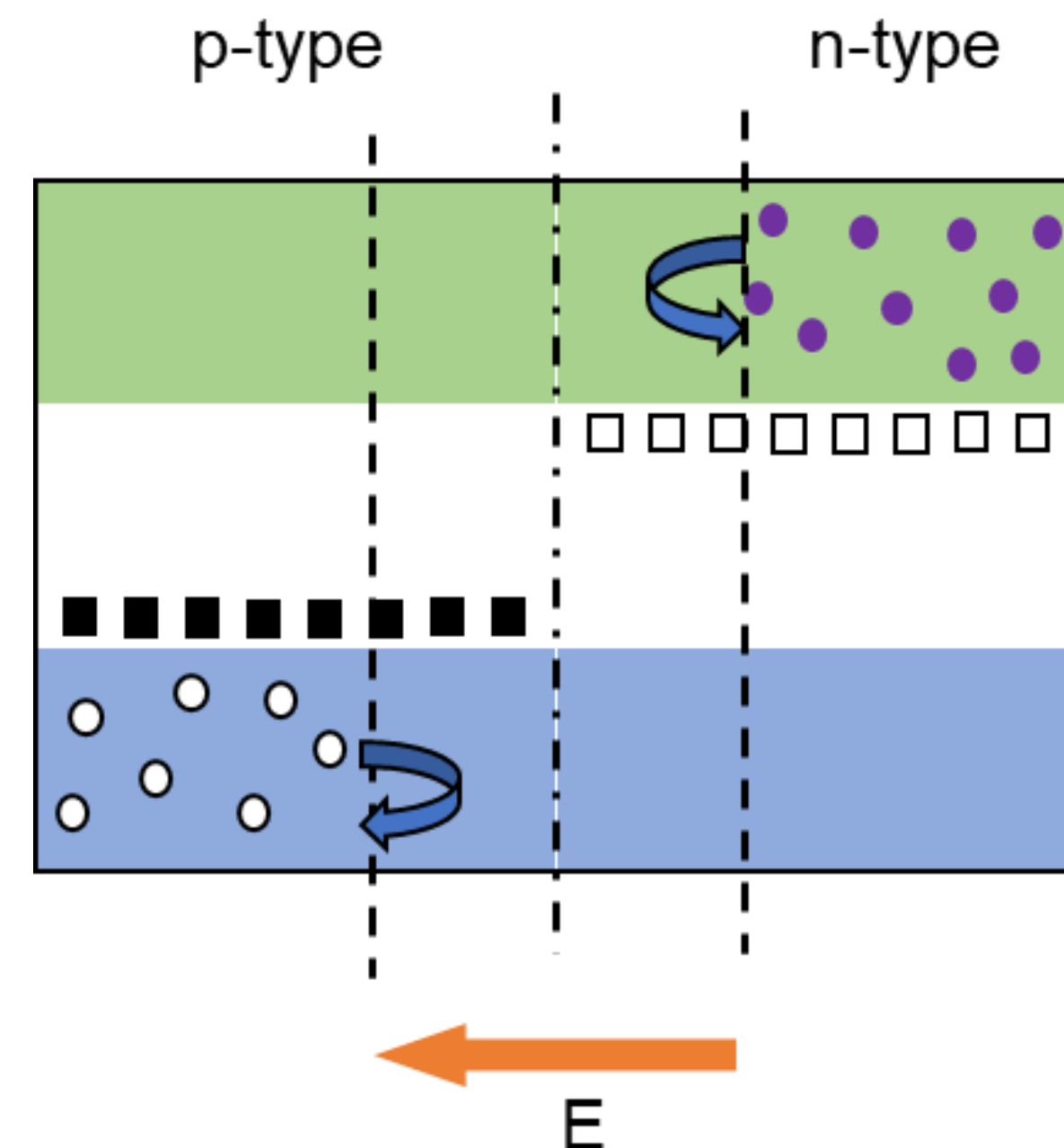
Free electrons on the n-side and free holes on the p-side can initially wander across the junction. When a free electron meets a free hole, it can “drop into it”. So far as charge movements are concerned, this means the hole and electron cancel each other and vanish.



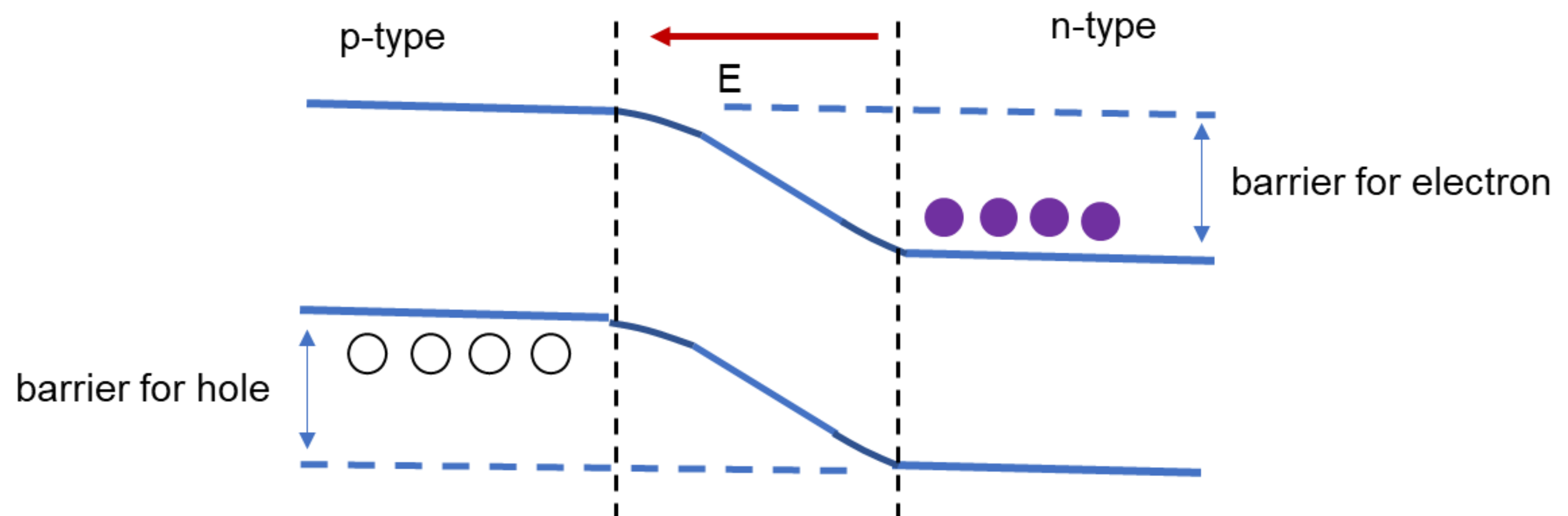
As a result, the free electrons and holes near the junction tend to eat each other, producing a region depleted of any moving charges. This creates what is called the depletion zone or space charge zone.



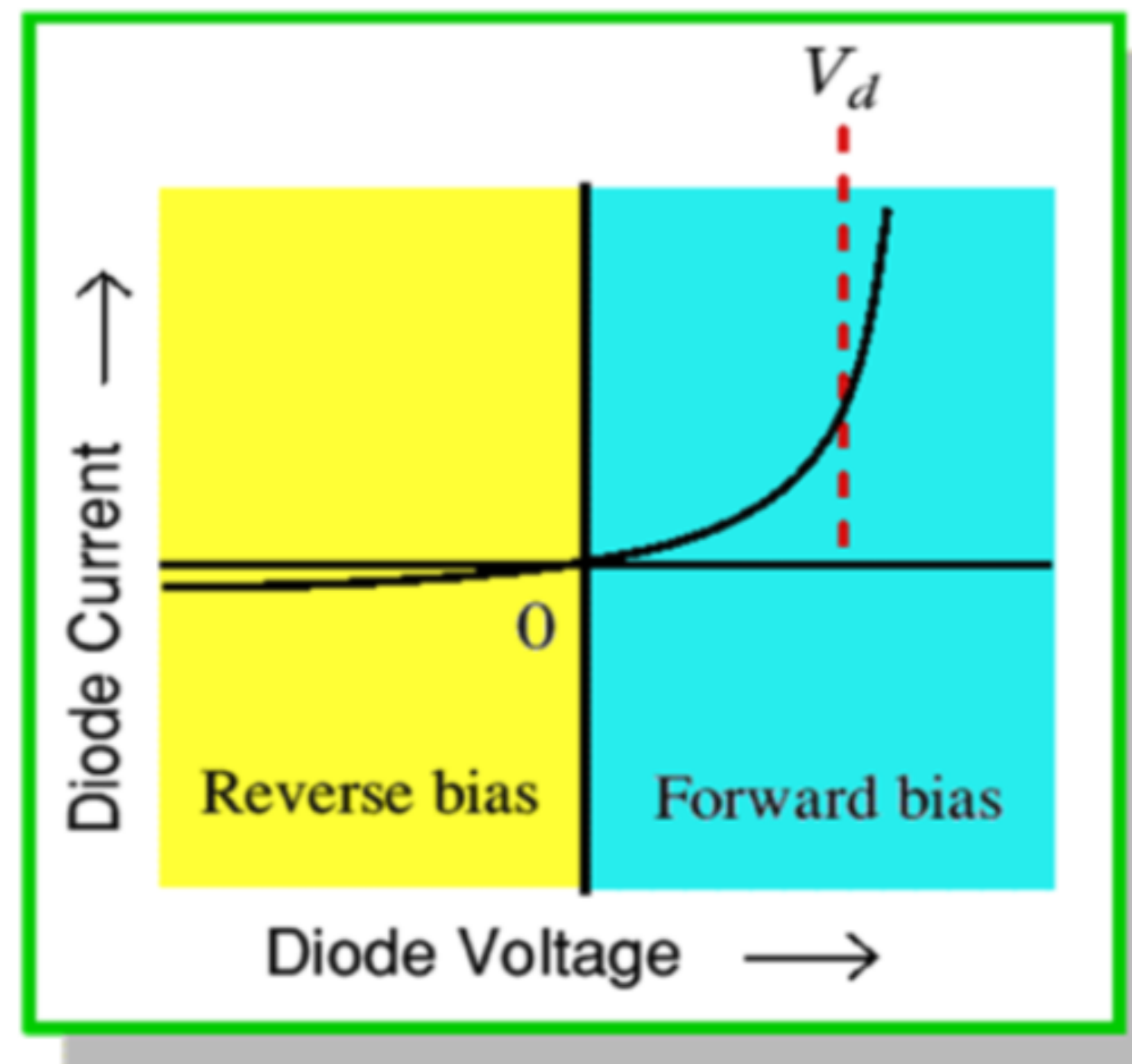
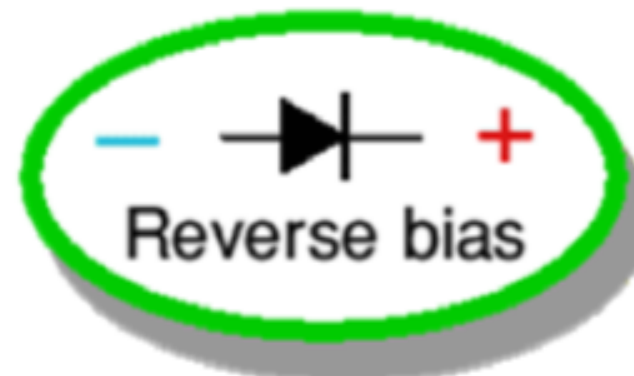
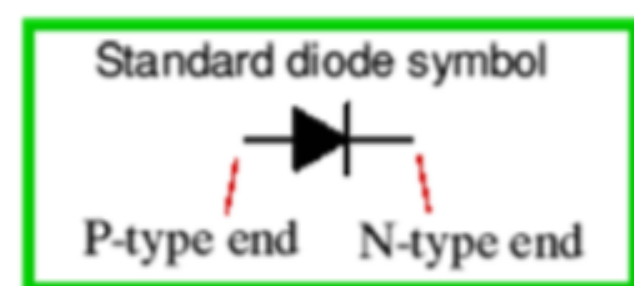
Now, any free charge which wanders into the depletion zone finds itself in a region with no other free charges. Locally it sees a lot of positive charges (the donor atoms) on the n-type side and a lot of negative charges (the acceptor atoms) on the p-type side. These exert a force on the free charge, driving it back to its “own side” of the junction away from the depletion zone.



- The acceptor and donor atoms are “nailed down” in the solid and cannot move around.
- However, the negative charge of the acceptor’s extra electron and positive of the donor’s extra proton (exposed by it’s missing electron) tend to keep the depletion zone swept clean of free charges once the zone has formed.
- A free charge now requires some extra energy to overcome the forces from the ionized donor/acceptor atoms to be able to cross the zone. The junction therefore acts like a barrier, blocking any charge flow (current) across the barrier.
- Usually, we represent this barrier by bending the conduction and valence bands as they cross the depletion zone.
- Now we can imagine, the electrons having to “get uphill” to move from the n-type side to the p-type side.
- The holes behave a bit like balloons bobbing up against a ceiling. On this kind of diagram, you require energy to “pull them down” before they can move from the p-side to the n-type side.
- The energy required by the free holes and electrons (to move to other side) can be supplied by a suitable voltage applied between the two ends of the p-n junction diode.



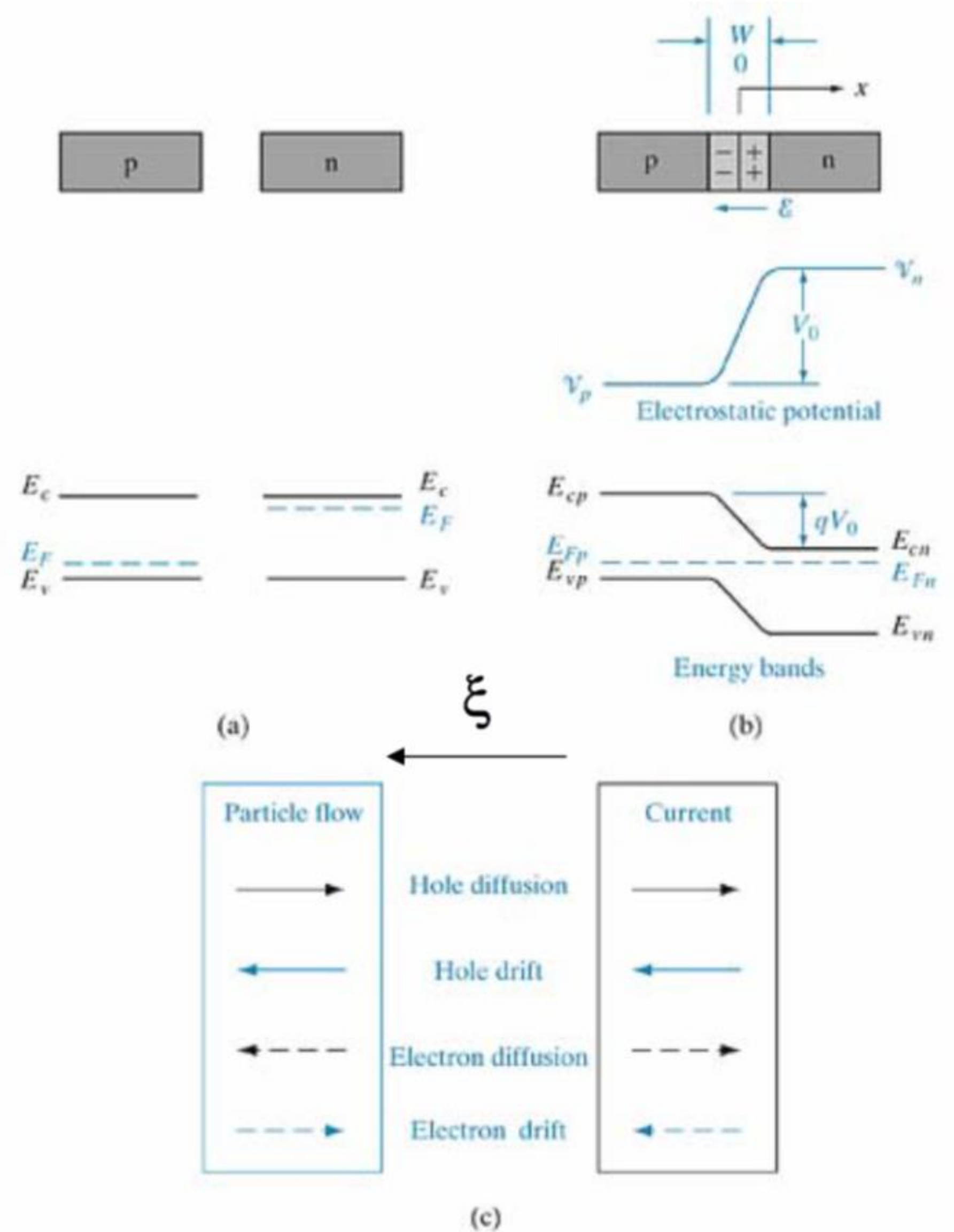
- Notice that this voltage must be supplied the correct way, around, this pushes the charges over the barrier (p-n junction forward bias)
- However, applying the voltage the “wrong” way around makes things worse by pulling what free charges there are away from the junction (p-n junction is in reverse bias)!
- This is why p-n junction (or diode) conducts in one direction but not the other.



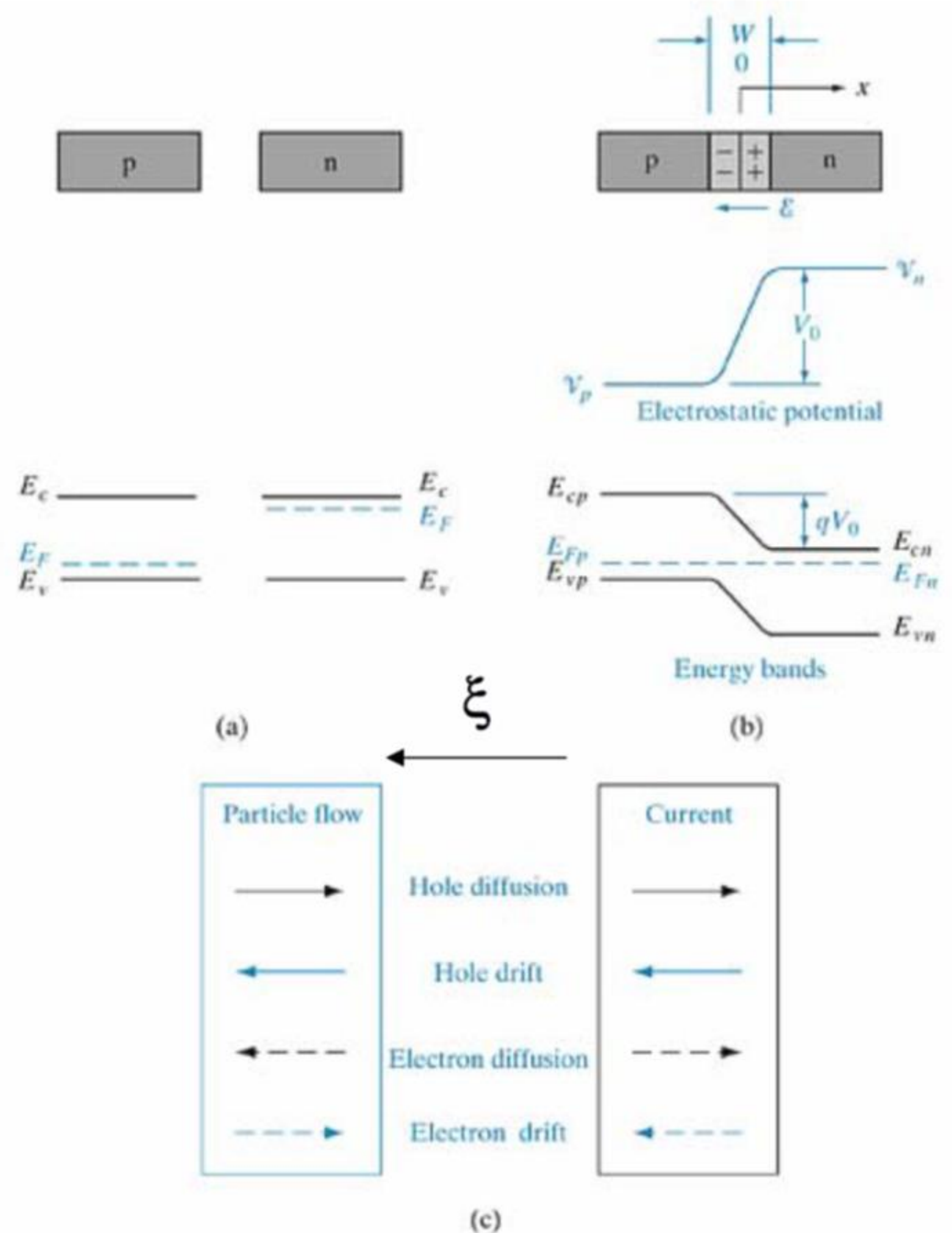
Forward bias - all current, almost no volts.
Reverse bias - all volts, almost no current.

Upon the formation of p-n junction

- Hole diffuse from p-side to n-side and electron diffuse from n-side to p-side.
- The diffusion of holes and electrons give rise to corresponding diffusion currents, $J_p(\text{diff})$ and $J_n(\text{diff})$, respectively.
- Both $J_p(\text{diff})$ and $J_n(\text{diff})$ flow from p-side to n-side.
- The diffusion current cannot build up indefinitely because of an opposing electric field that is created at the junction.

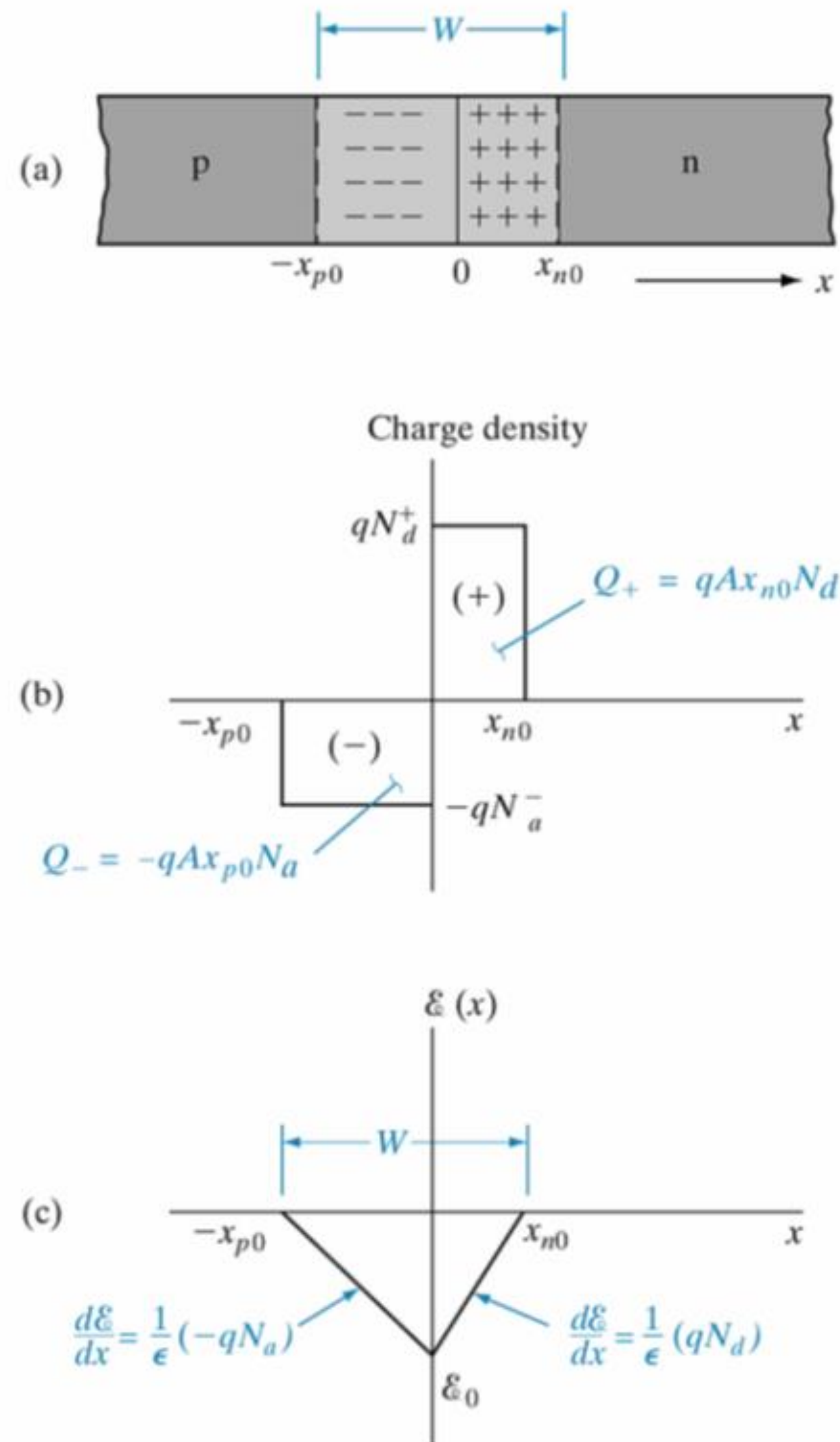


- Electrons diffusion from n to p leave behind uncompensated (ionized) donor ions N_d^+ in the n-material.
- Holes diffusion from p to n leave behind uncompensated (ionized) acceptor ions N_a^- in the p-material.
- The resulting electric field (E) is directed from n-side to p-side.
- The electric field E is in a direction opposite to that of diffusion current of both electrons and holes – lowers diffusion current.
- The electric field E leads to drift currents ($J_{drift} = \sigma E$)
- $J_p(\text{drift})$ and $J_n(\text{drift})$ flow from n-side to p-side, in the direction of the electric field E.



- At equilibrium, there is no net current flow across the p-n junction.
- This means the drift current due to electric field E must exactly balance the diffusion current.
- Also, since there is no net build up of either electrons or holes on either side as a function of time, the drift and diffusion currents must cancel for each type of carrier.
- $J_p = J_p(\text{drift}) + J_p(\text{diff}) = 0$
- $J_n = J_n(\text{drift}) + J_n(\text{diff}) = 0$
- The total current density $J = J_p + J_n = 0$, at thermal equilibrium.

- The electric field E thus would have to build up to the point where the net current J is zero at equilibrium.
- This electric field E appears across a region W about the junction, and there is an equilibrium potential difference V_o across W .
- The electric field is zero in the neutral regions of the semiconductor outside W – implies a constant potential V_n in the neutral n-material and a constant potential V_p in the neutral p-material.
- There exists thus, a potential difference $V_o = V_n - V_p$ over a region of width W .
- The region W is called the depletion region (also called the transition or space-charge region) and the potential difference V_o is called the contact potential (also called built-in potential).
- Note that the contact potential appearing across the depletion region is a built-in barrier and not an external potential. Thus, the contact potential cannot be measured by placing a voltmeter across a p-n junction.



Space charge and electric field distribution within the transition region of a p-n junction with $N_d > N_a$: (a) the transition region, with $x = 0$ defined at the metallurgical junction; (b) charge density within the transition region, neglecting the free carriers; (c) the electric field distribution, where the reference direction for ξ is arbitrarily taken as the +x-direction.

Thermal equilibrium condition

At equilibrium condition, the drift current due to the electric field must exactly cancel the diffusion current due to the concentration gradient.

$$J_n = e\mu_n n \mathbf{E} + eD_n \frac{dn}{dx} = 0$$

$$J_p = e\mu_p p \mathbf{E} - eD_p \frac{dp}{dx} = 0$$

$$\therefore \frac{dE_F}{dx} = 0$$

the Fermi levels in the p- and n-type semiconductors must be equal

When a semiconductor is in equilibrium, there's no *net* flux

$$\text{Since } F_n = D_n \frac{dn}{dx} + \mu_n n \mathcal{E} = 0$$

$$n = n_i e^{(E_F - E_i)/kT}$$

$$\frac{dn}{dx} = \frac{n}{kT} \left[\frac{dE_F}{dx} - \frac{dE_i}{dx} \right]$$

Then

$$F_n = \frac{D_n}{kT} n \left[\frac{dE_F}{dx} - \frac{dE_i}{dx} \right] + \mu_n n \frac{1}{q} \frac{dE_i}{dx}$$

Using the Einstein relationship,

$$\frac{D_n}{kT} = \frac{\mu_n}{q}$$

$$F_n = \frac{D_n}{kT} n \frac{dE_F}{dx} \quad \text{A gradient in the Fermi level is the driving force for carrier motion}$$

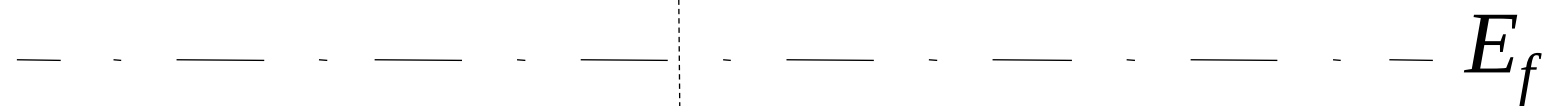
In equilibrium, $F_n = 0 \Rightarrow E_F = \text{constant}$.

This is an extremely important result!

Energy Band Diagram of a PN Junction

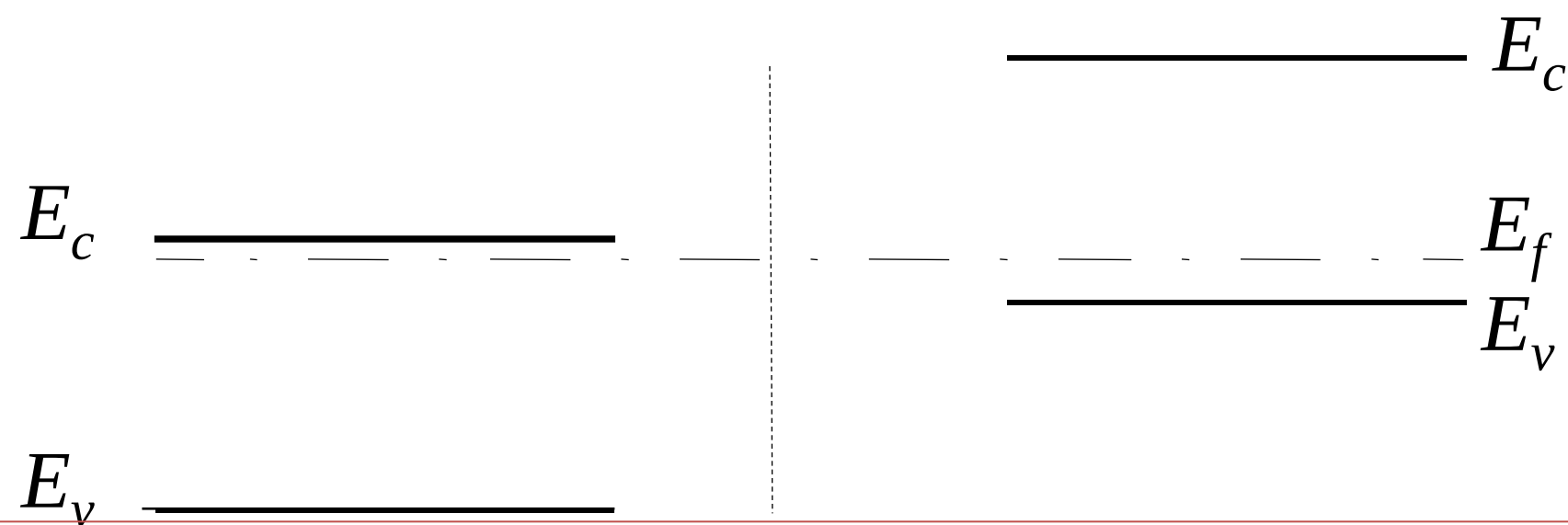
N-region ← → **P-region**

(a)



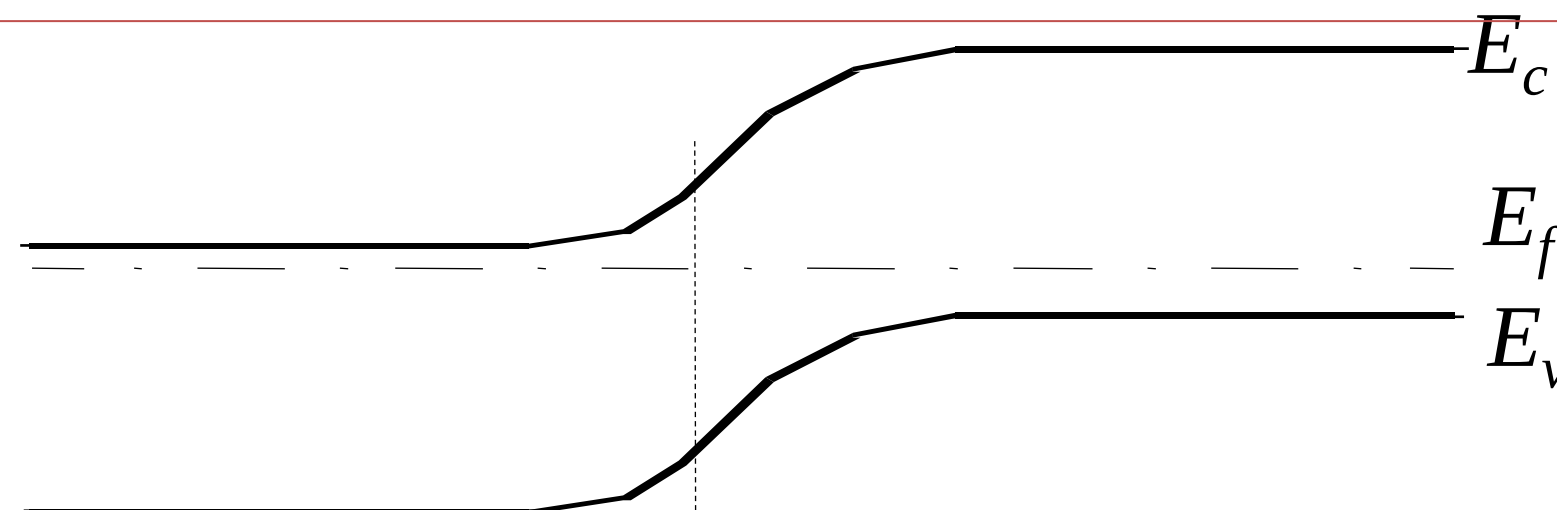
E_f is constant at equilibrium

(b)



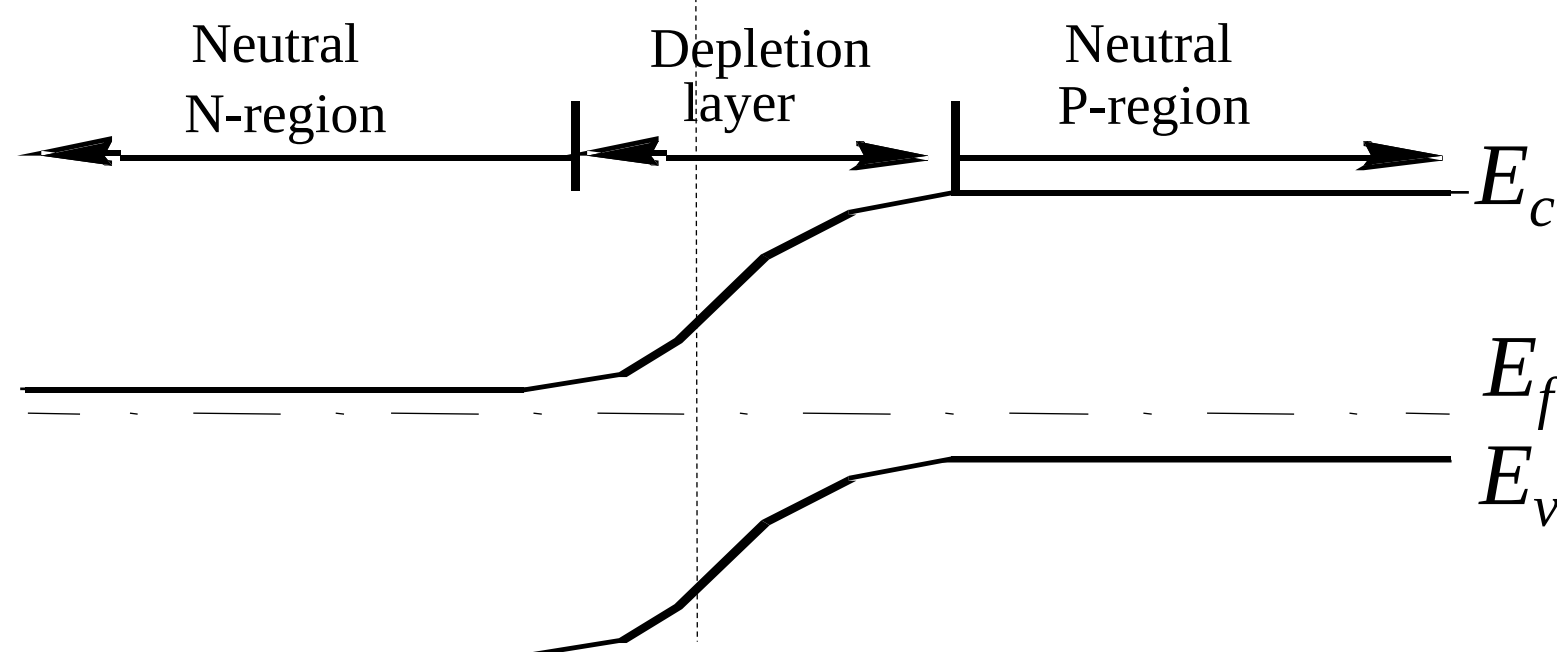
E_c and E_v are known relative to E_f

(c)



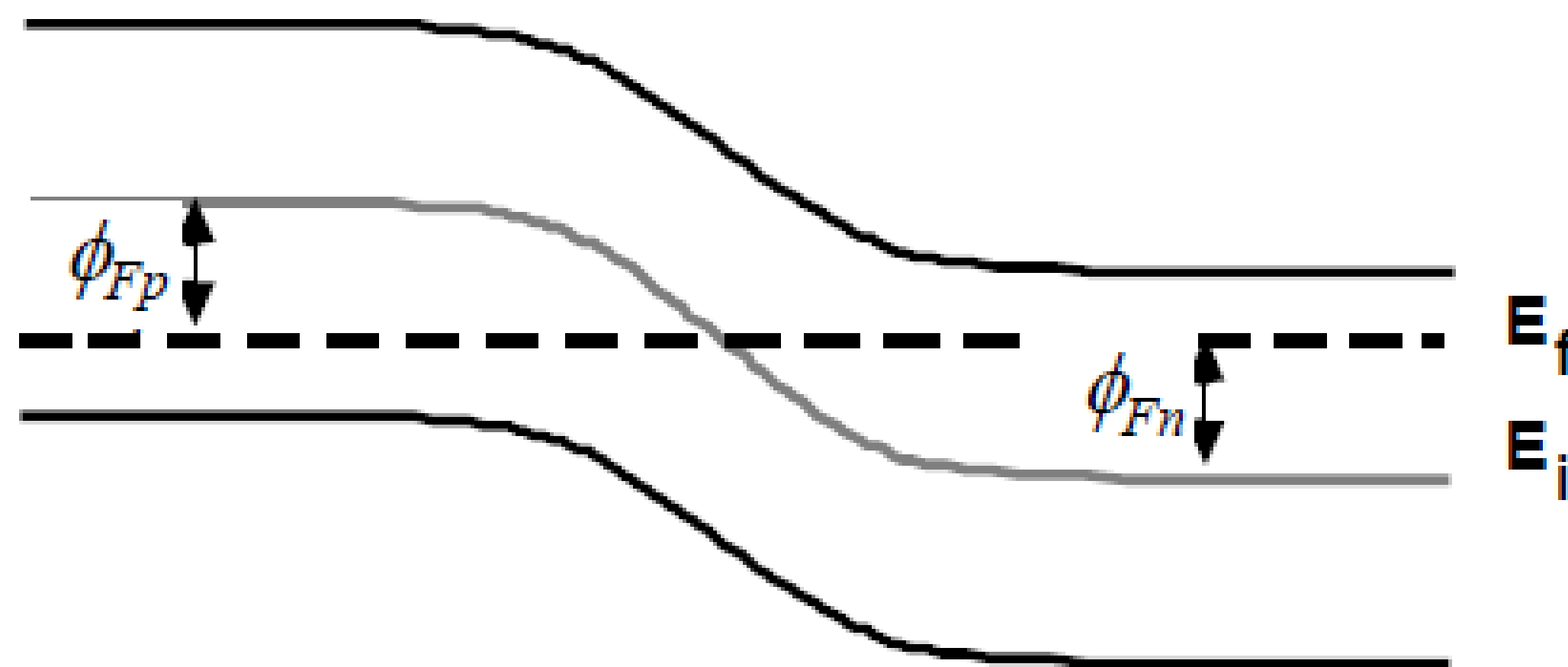
E_c and E_v are smooth, the exact shape to be determined.

(d)



A depletion layer exists at the PN junction where $n \approx 0$ and $p \approx 0$.

Energy Bands at Equilibrium



Thus, with constant Fermi level,

$$V_{bi} = \phi_{Fn} - \phi_{Fp} = \text{built in potential}$$

$$\begin{aligned} &= \frac{kT}{q} \left[\ln \frac{N_D}{n_i} + \ln \frac{N_A}{n_i} \right] \\ &= \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2} \end{aligned} \quad (3)$$

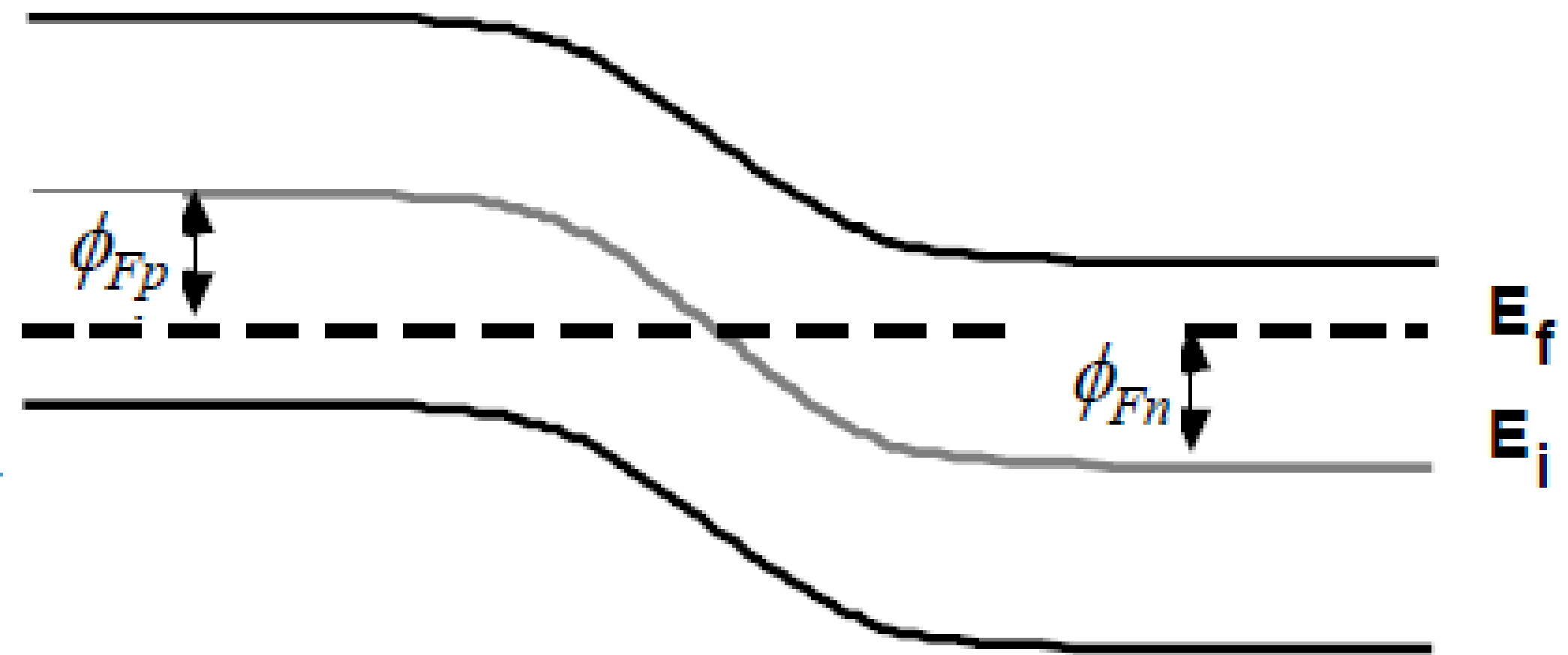
The built in potential is only dependent upon the carrier densities, not diffusion coefficient, lifetime, etc. The higher the doping, the larger the built in potential.

V_{bi} : Quiz

$$V_{bi} = \phi_{Fn} - \phi_{Fp} = \text{built in potential}$$

$$= \frac{kT}{q} \left[\ln \frac{N_D}{n_i} + \ln \frac{N_A}{n_i} \right]$$

$$= \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$



- What is the maximum value of V_{bi} ?
- Does V_{bi} increase / decrease with temperature ?

1D Poisson's equation:

$$E(x) = -\frac{d\psi}{dx}$$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{d\mathbf{E}(x)}{dx} = -\frac{\rho_s(x)}{\varepsilon} =$$

$$= -\frac{e}{\varepsilon} [N_D(x) - N_A(x) + p(x) - n(x)]$$

ψ - electrostatical potential

ρ - space charge density

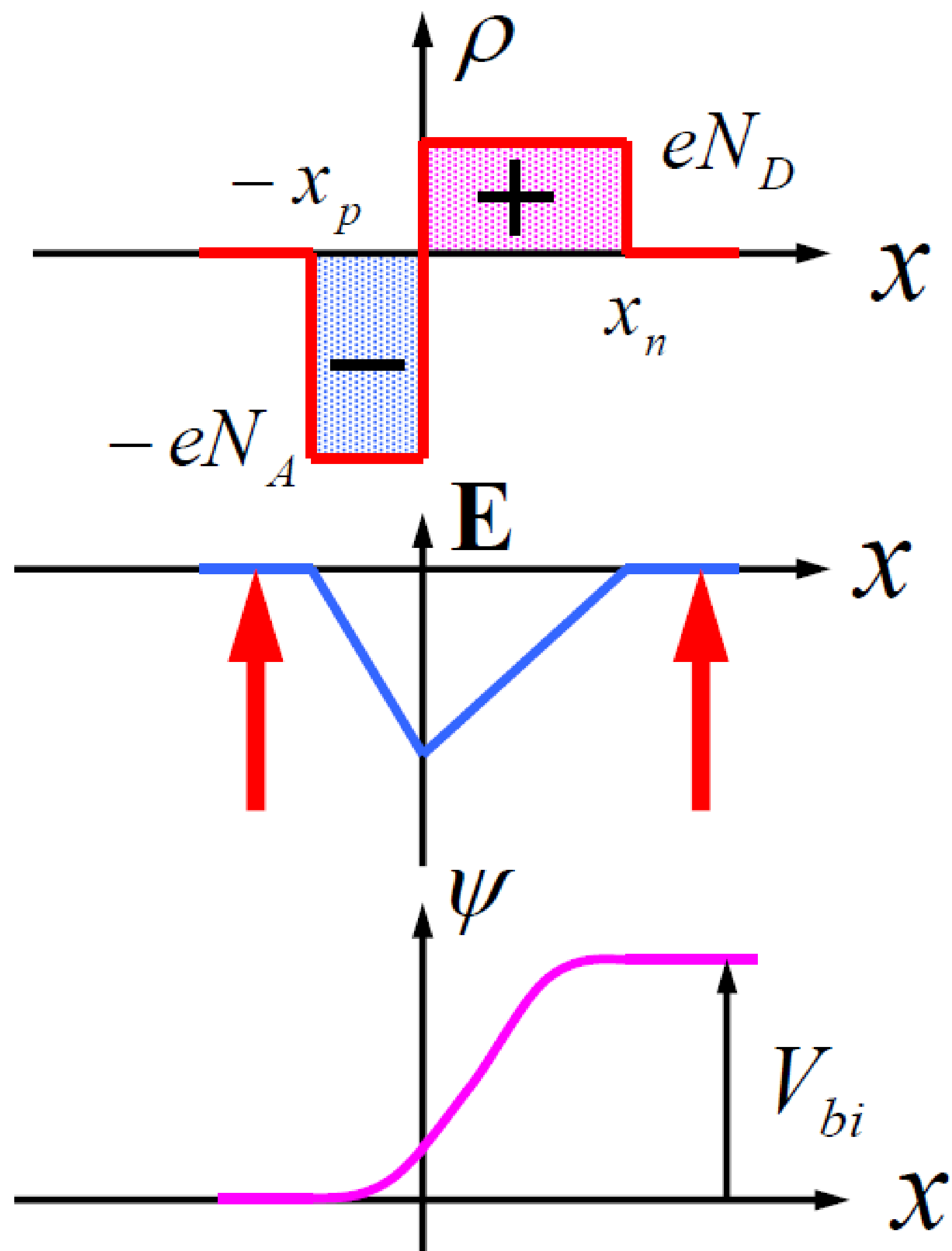
ε - semiconductor permittivity

Poisson's equation for abrupt junction

$$\frac{d^2\psi(x)}{dx^2} = -\frac{d\mathbf{E}(x)}{dx} = \frac{eN_A}{\varepsilon} \quad \text{for} \quad -x_p \leq x < 0$$

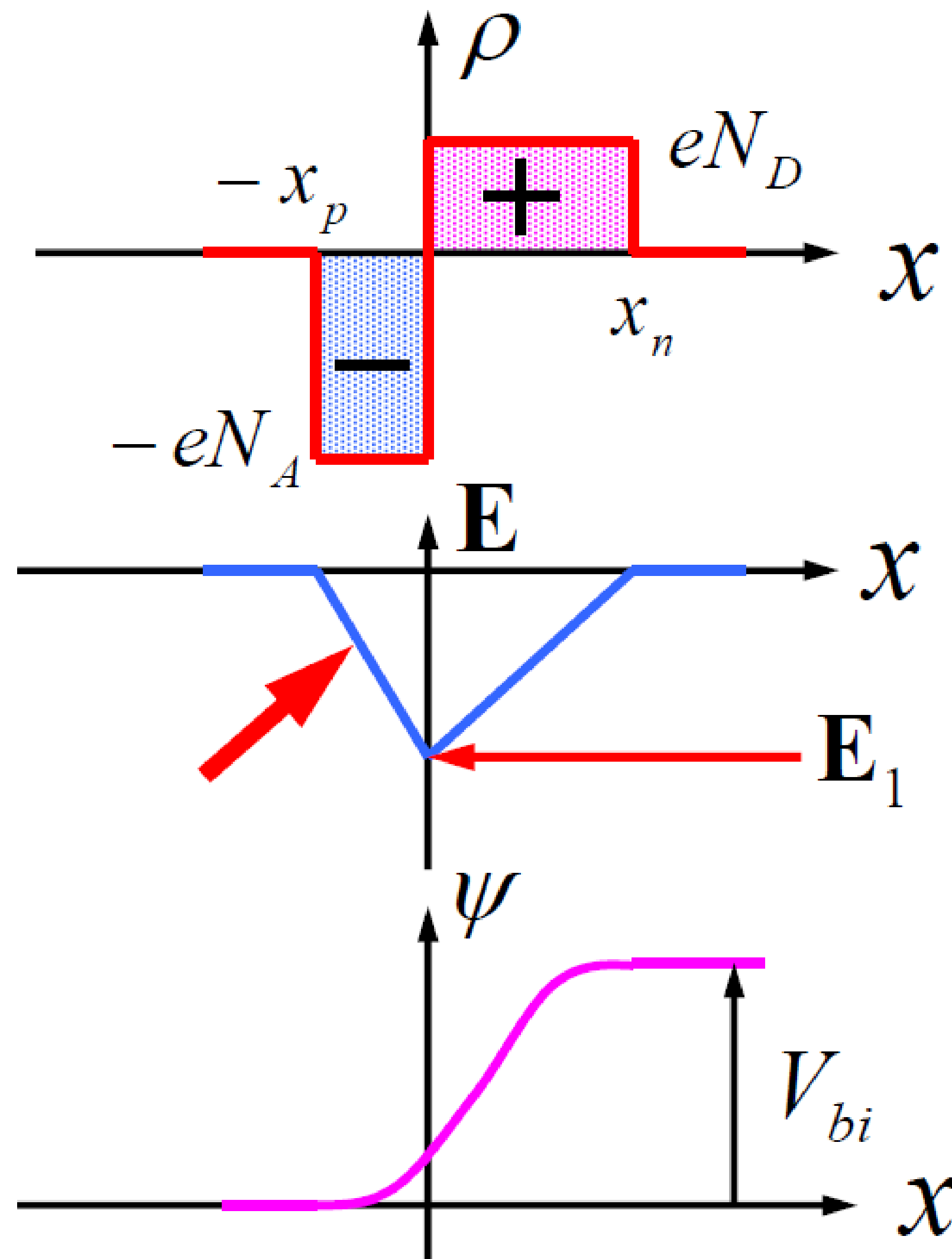
$$\frac{d^2\psi(x)}{dx^2} = -\frac{d\mathbf{E}(x)}{dx} = \frac{-eN_D}{\varepsilon} \quad \text{for} \quad 0 < x \leq x_n$$

Junction potential



$$\frac{d\mathbf{E}(x)}{dx} = 0$$

Electric field distribution



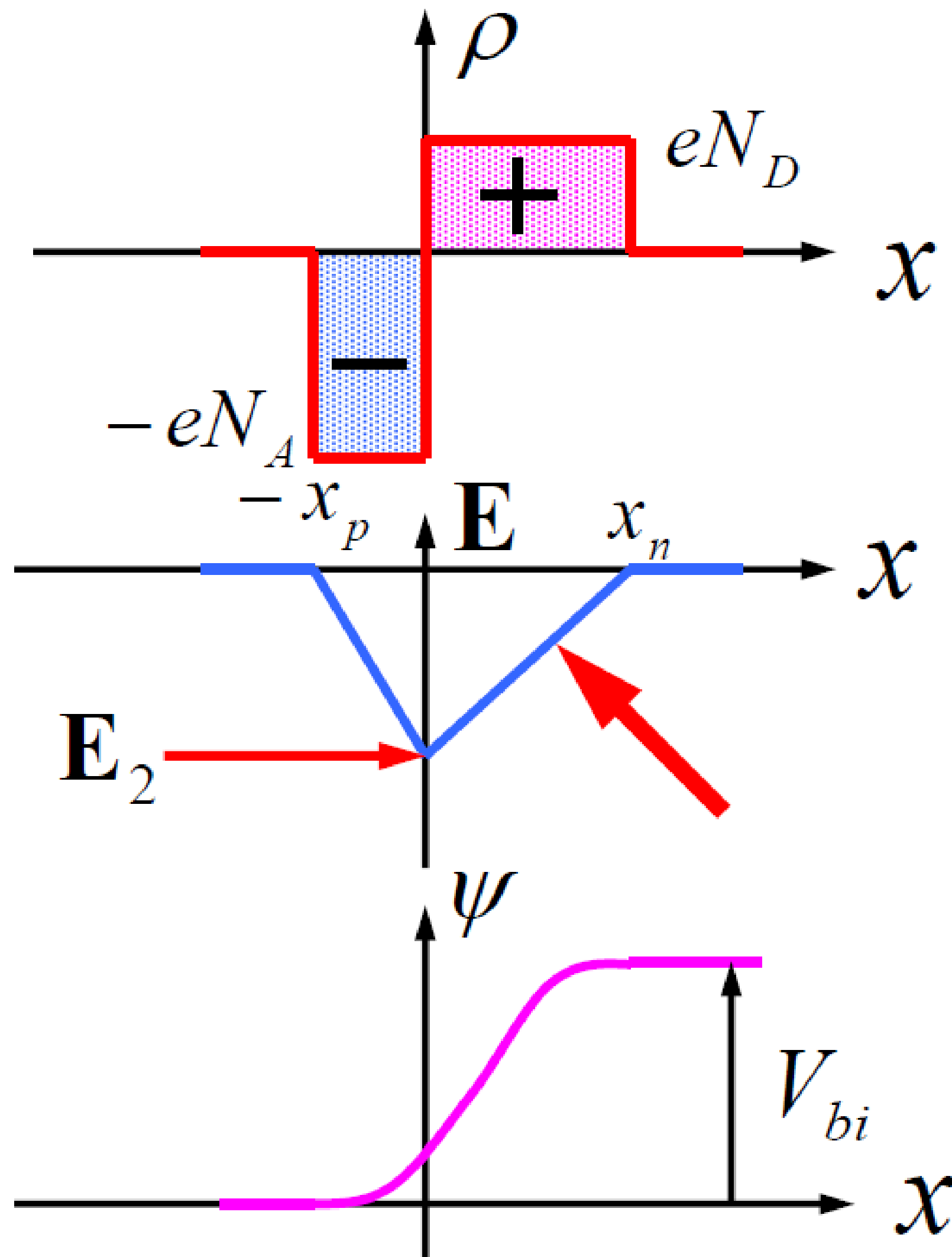
$$\frac{d\mathbf{E}(x)}{dx} = \frac{-eN_A}{\epsilon}$$

$$\mathbf{E}(x) = -\frac{eN_A}{\epsilon}x + \mathbf{E}_1$$

$$\mathbf{E}_1 = -\frac{eN_A}{\epsilon}x_p$$

$$\mathbf{E}(x) = -\frac{eN_A}{\epsilon}(x + x_p)$$

Electric field distribution



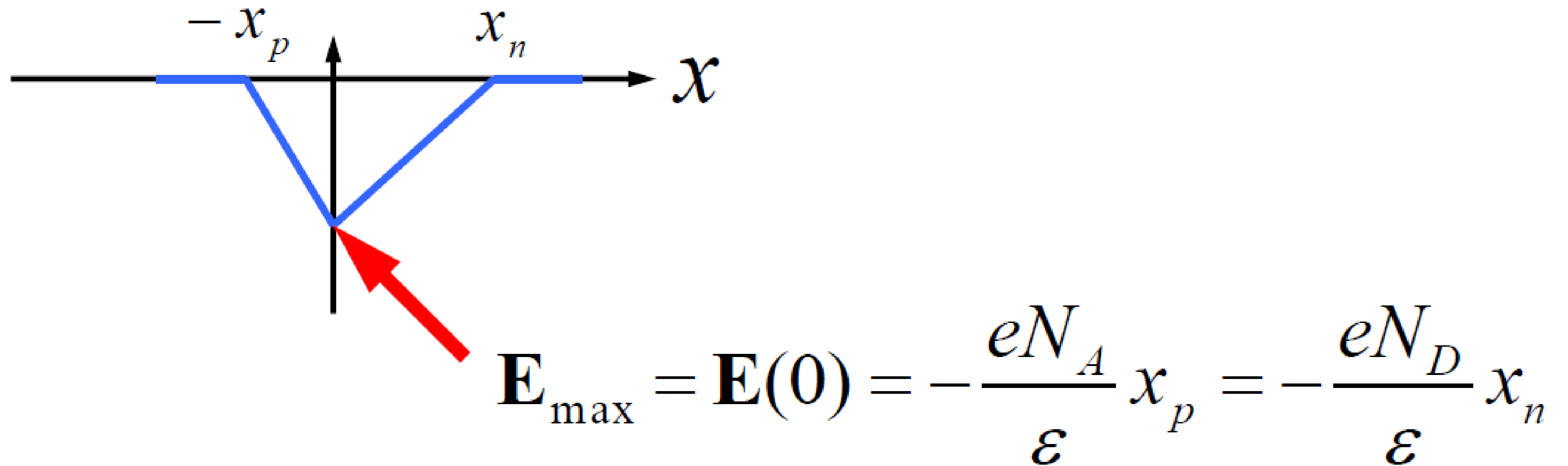
$$\frac{d\mathbf{E}(x)}{dx} = \frac{eN_D}{\epsilon}$$

$$\mathbf{E}(x) = \frac{eN_D}{\epsilon} x + \mathbf{E}_2$$

$$\mathbf{E}_2 = -\frac{eN_D}{\epsilon} x_n$$

$$\mathbf{E}(x) = \frac{eN_D}{\epsilon} (x - x_n)$$

Maximum electric field



consequence:

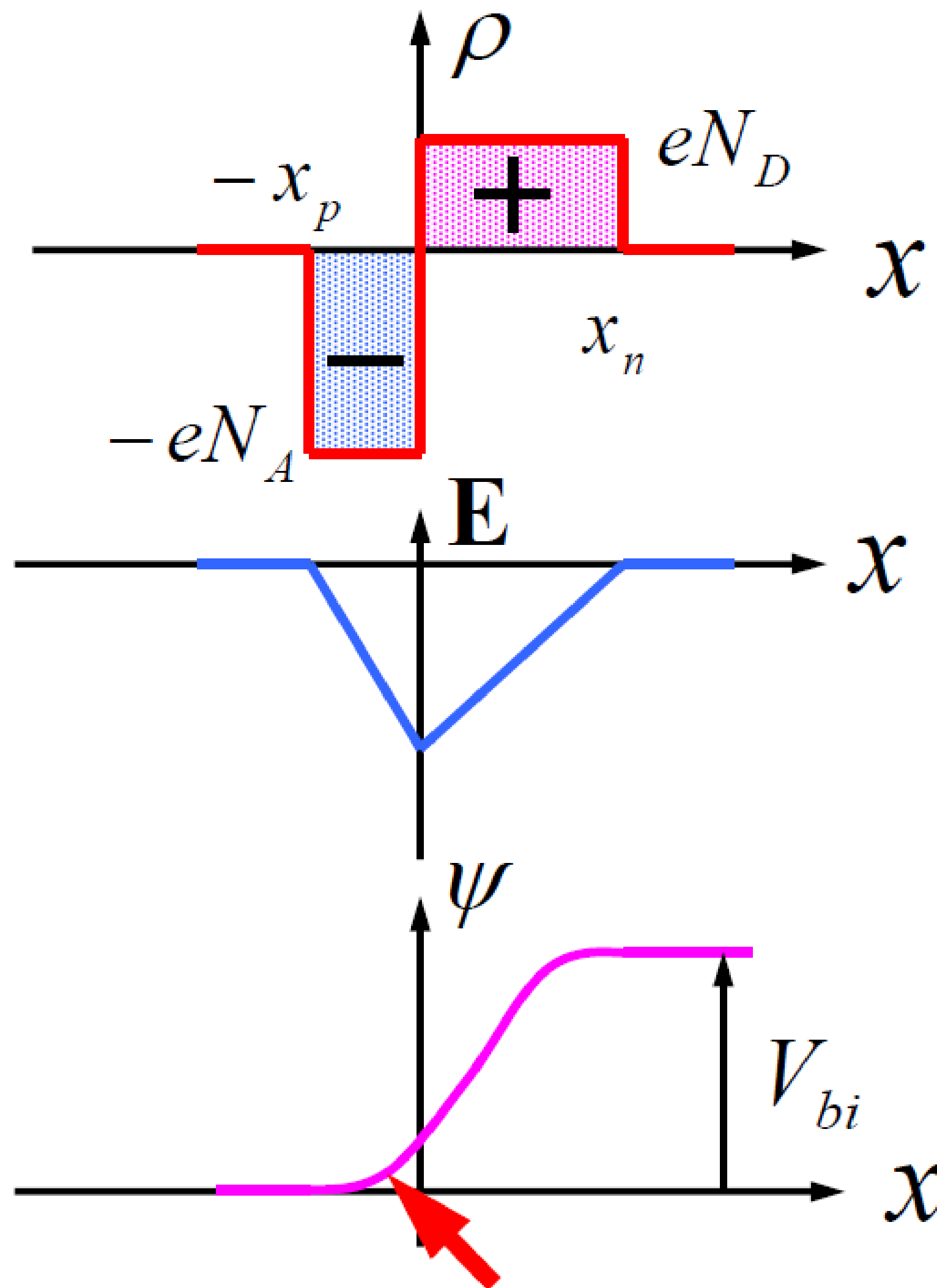
$$N_A x_p = N_D x_n$$

Potential distribution

$$\mathbf{E}(x) = -\frac{d\psi(s)}{dx}$$

$$\psi(s) = -\int \mathbf{E}(x)dx$$

Potential distribution



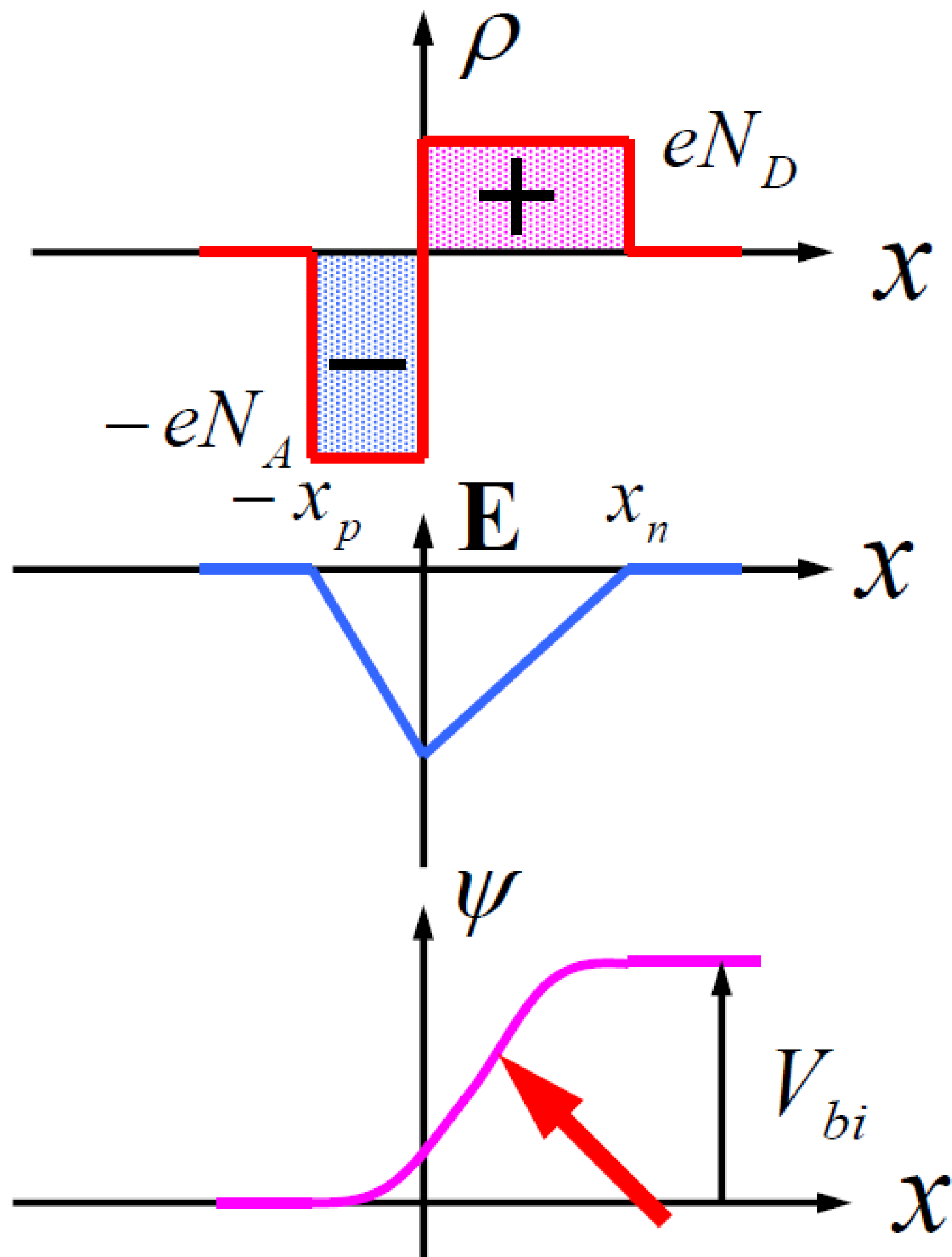
$$\begin{aligned}\psi(x) &= \int \frac{eN_A}{\varepsilon} (x + x_p) dx \\ &= \frac{eN_A}{\varepsilon} \left(\frac{x^2}{2} + x_p x \right) + \psi_1\end{aligned}$$

with $\psi(-x_p) = 0$

$$\psi_1 = \frac{eN_A}{\varepsilon} \frac{x_p^2}{2}$$

$$\psi(x) = \frac{eN_A}{2\varepsilon} (x_p + x)^2$$

Potential distribution



$$\begin{aligned}\psi(x) &= \int \frac{eN_D}{\varepsilon} (x_n - x) dx \\ &= \frac{eN_D}{\varepsilon} \left(x_n x - \frac{x^2}{2} \right) + \psi_2\end{aligned}$$

with $\psi(x_n) = V_{bi}$

$$\psi_2 = V_{bi} - \frac{eN_D}{\varepsilon} \frac{x_n^2}{2}$$

$$\psi(x) = V_{bi} - \frac{eN_D}{2\varepsilon} (x_n - x)^2$$

Built-in potential

for $x = 0$ both expressions

$$\psi(x) = V_{bi} - \frac{eN_D}{2\epsilon} (x_n - x)^2 \qquad \psi(x) = \frac{eN_A}{2\epsilon} (x_p + x)^2$$

must give the same value:

$$\psi(0) = V_{bi} - \frac{eN_D}{2\epsilon} x_n^2 = \frac{eN_A}{2\epsilon} x_p^2$$

$$V_{bi} = \frac{e}{2\epsilon} (N_D x_n^2 + N_A x_p^2)$$

Depletion width

$$x_d \equiv x_n + x_p$$

$$V_{bi} = \frac{e}{2\varepsilon} \left(N_D x_n^2 + N_A x_p^2 \right) \quad N_A x_p = N_D x_n$$

$$V_{bi} = \frac{e}{2\varepsilon} \left[N_D x_n^2 + N_A \left(\frac{N_D x_n}{N_A} \right)^2 \right] = x_n^2 \frac{e}{2\varepsilon} \left(N_D + \frac{N_D^2}{N_A} \right)$$

$$V_{bi} = \frac{e}{2\varepsilon} \left[N_D \left(\frac{N_A x_p}{N_D} \right)^2 + N_A x_p^2 \right] = x_p^2 \frac{e}{2\varepsilon} \left(\frac{N_A^2}{N_D} + N_A \right)$$

$$x_n = \sqrt{\frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}}$$

$$x_p = \sqrt{\frac{2\varepsilon}{e} \frac{N_D}{N_D N_A + N_A^2} V_{bi}}$$

Depletion width

$$x_n = \sqrt{\frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}}$$

$$x_p = \sqrt{\frac{2\varepsilon}{e} \frac{N_D}{N_D N_A + N_A^2} V_{bi}}$$

$$x_d^2 = (x_n + x_p)^2 = \frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi} + \frac{2\varepsilon}{e} \frac{N_D}{N_A N_D + N_A^2} V_{bi}$$

$$+ 2 \cdot \sqrt{\frac{2\varepsilon}{e} \frac{N_A}{N_D N_A + N_D^2} V_{bi}} \cdot \sqrt{\frac{2\varepsilon}{e} \frac{N_D}{N_A N_D + N_A^2} V_{bi}} =$$

$$\frac{2\varepsilon}{e} V_{bi} \frac{N_A^2 + 2N_D N_A + N_D^2}{(N_A + N_D) N_D N_A} = \frac{2\varepsilon (N_A + N_D)}{e N_D N_A} V_{bi}$$

$$x_d = \sqrt{\frac{2\varepsilon (N_A + N_D)}{e N_D N_A} V_{bi}}$$

$$1. \quad V_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$

2.

$$x_d = \sqrt{\frac{2\varepsilon}{e} \frac{(N_A + N_D)}{N_D N_A} V_{bi}}$$

$$x_p = \frac{W N_D}{N_A + N_D} \quad x_n = \frac{W N_A}{N_A + N_D}$$

Note: $x_d = W =$ depletion width

3.

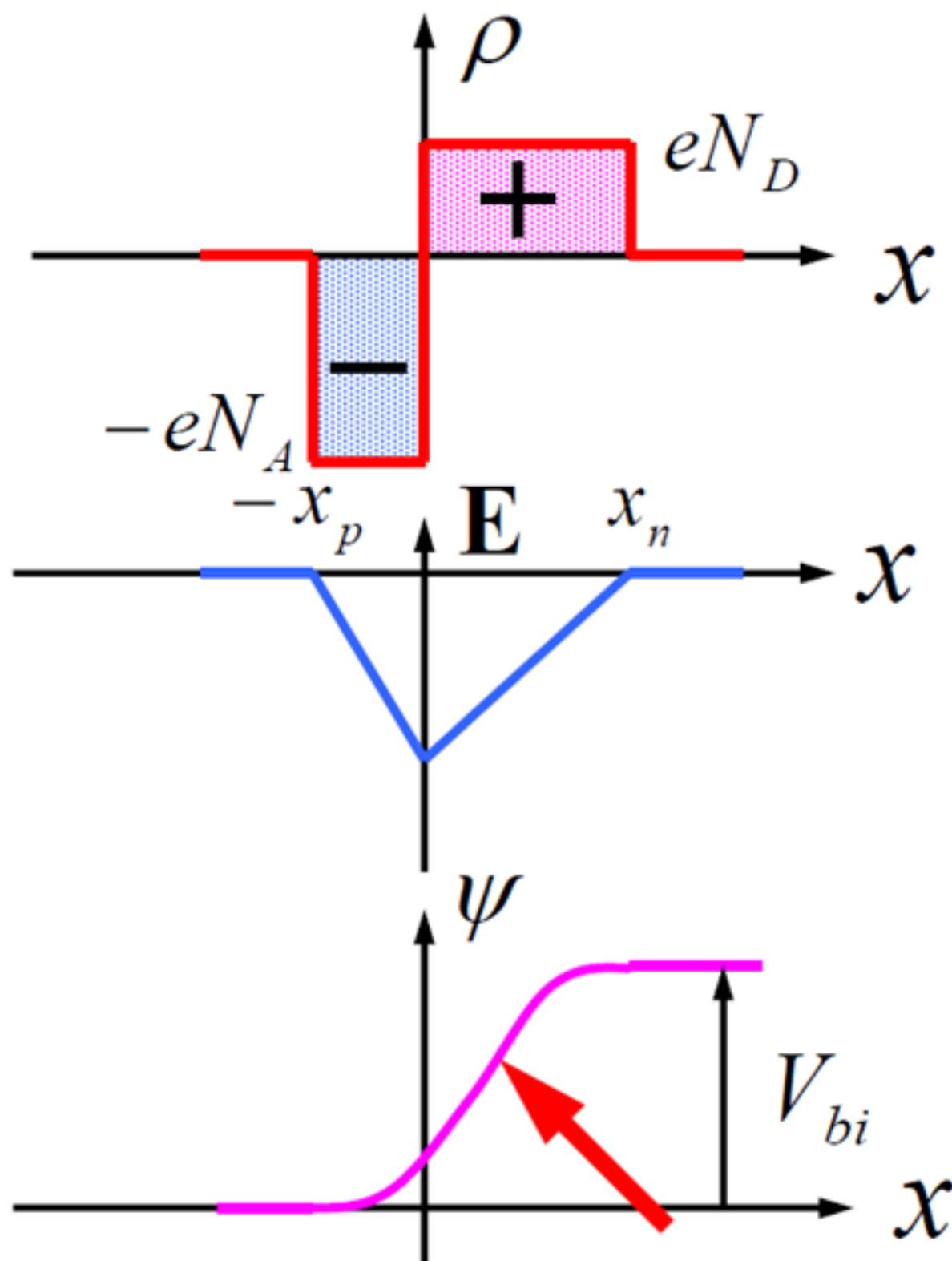
$$\mathbf{E}(x) = -\frac{eN_A}{\varepsilon} (x + x_p)$$

$$\mathbf{E}(x) = \frac{eN_D}{\varepsilon} (x - x_n)$$

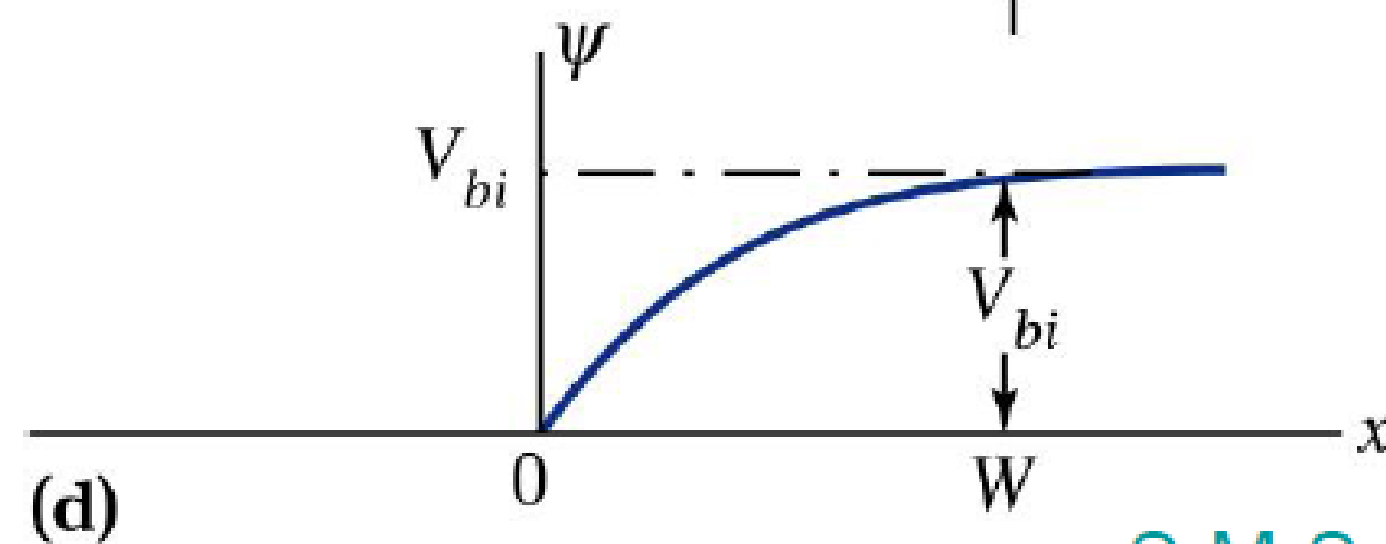
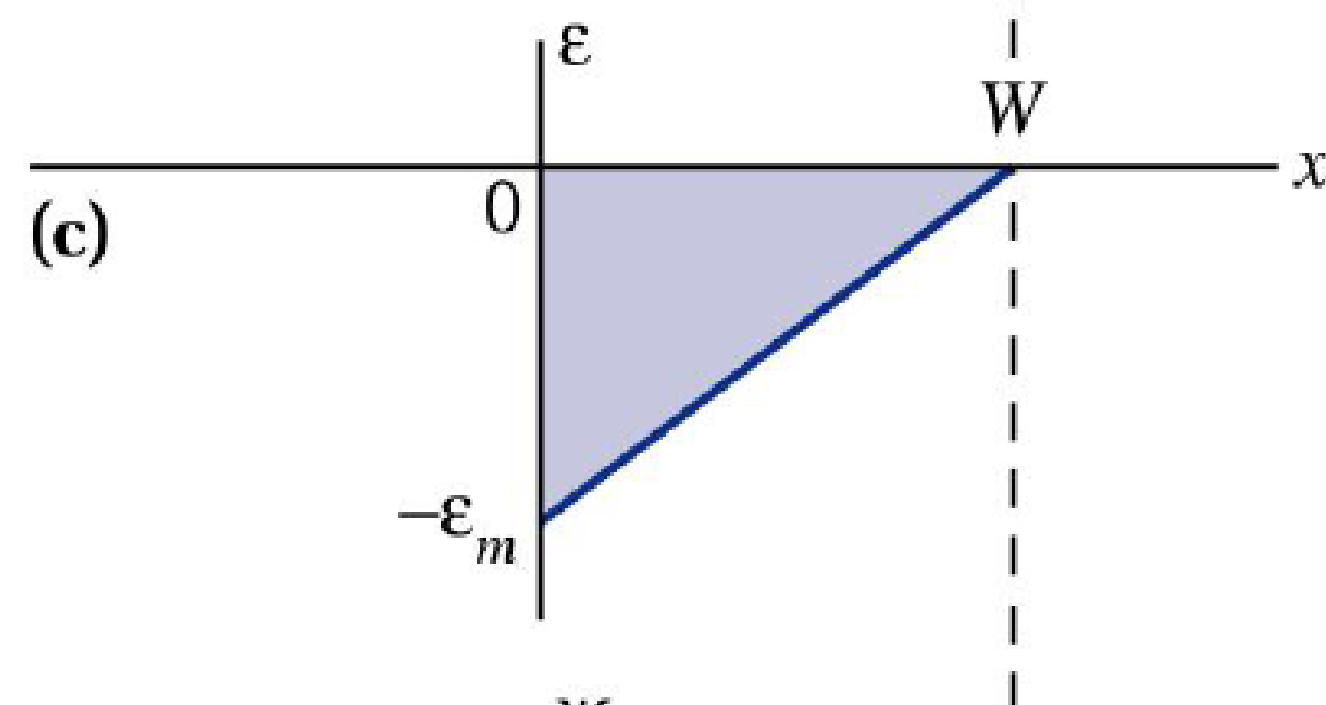
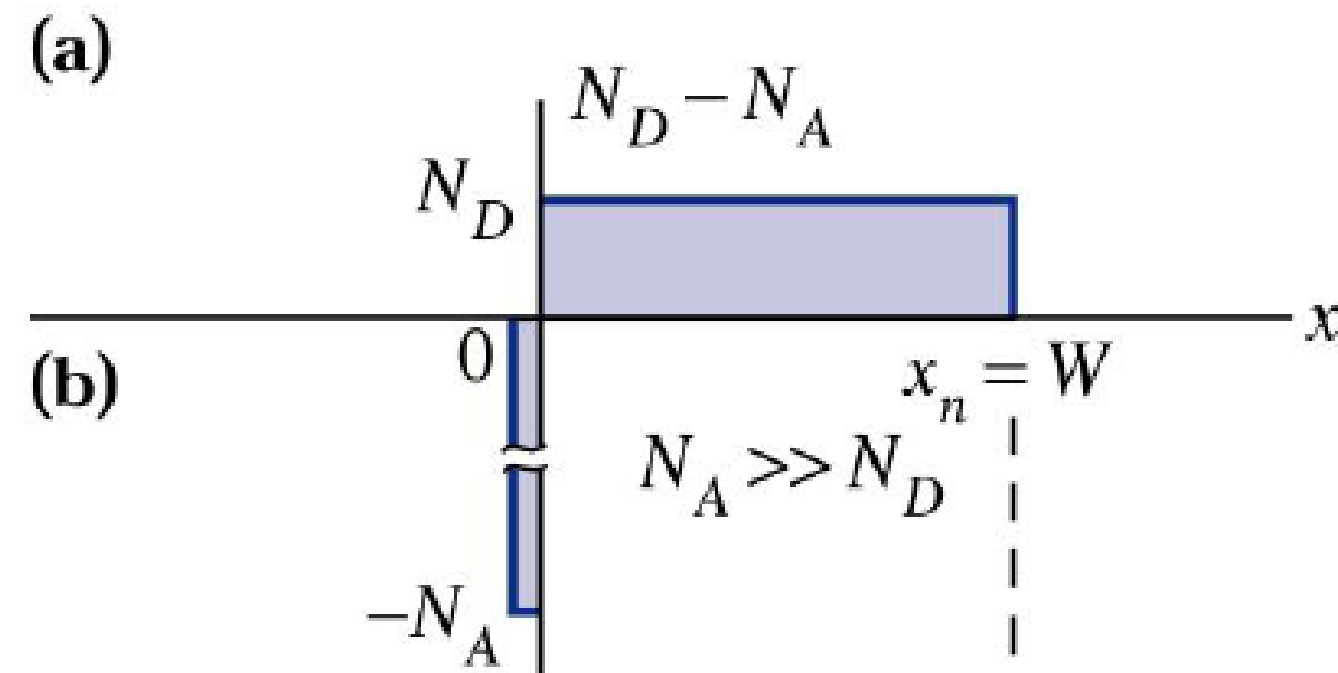
4.

$$\psi(x) = \frac{eN_A}{2\varepsilon} (x_p + x)^2$$

$$\psi(x) = V_{bi} - \frac{eN_D}{2\varepsilon} (x_n - x)^2$$



One-side abrupt junction



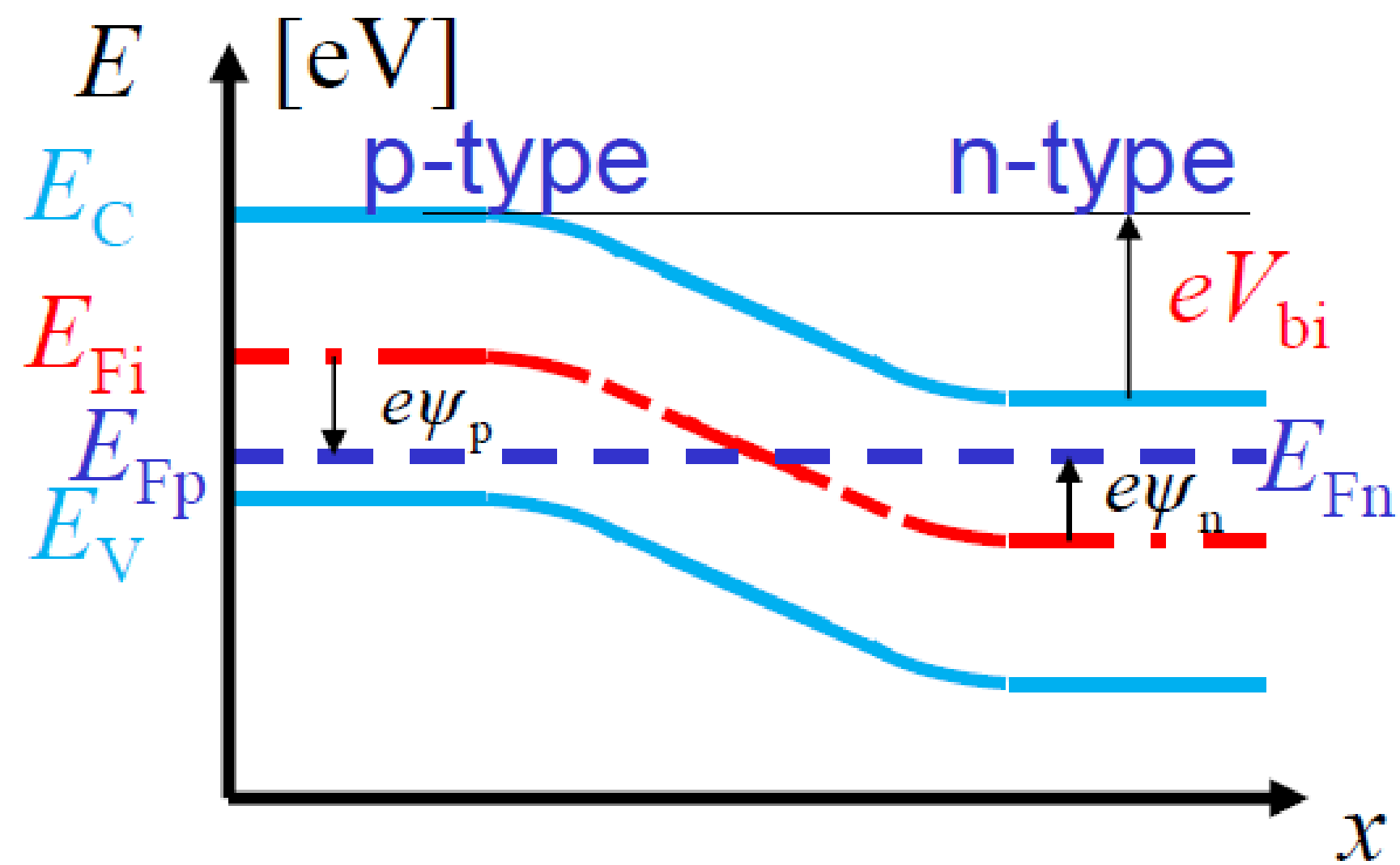
S.M.Sze

$$x_d = \sqrt{\frac{2\epsilon}{e} \frac{(N_A + N_D)}{N_D N_A} V_{bi}}$$

if $N_A \gg N_D$

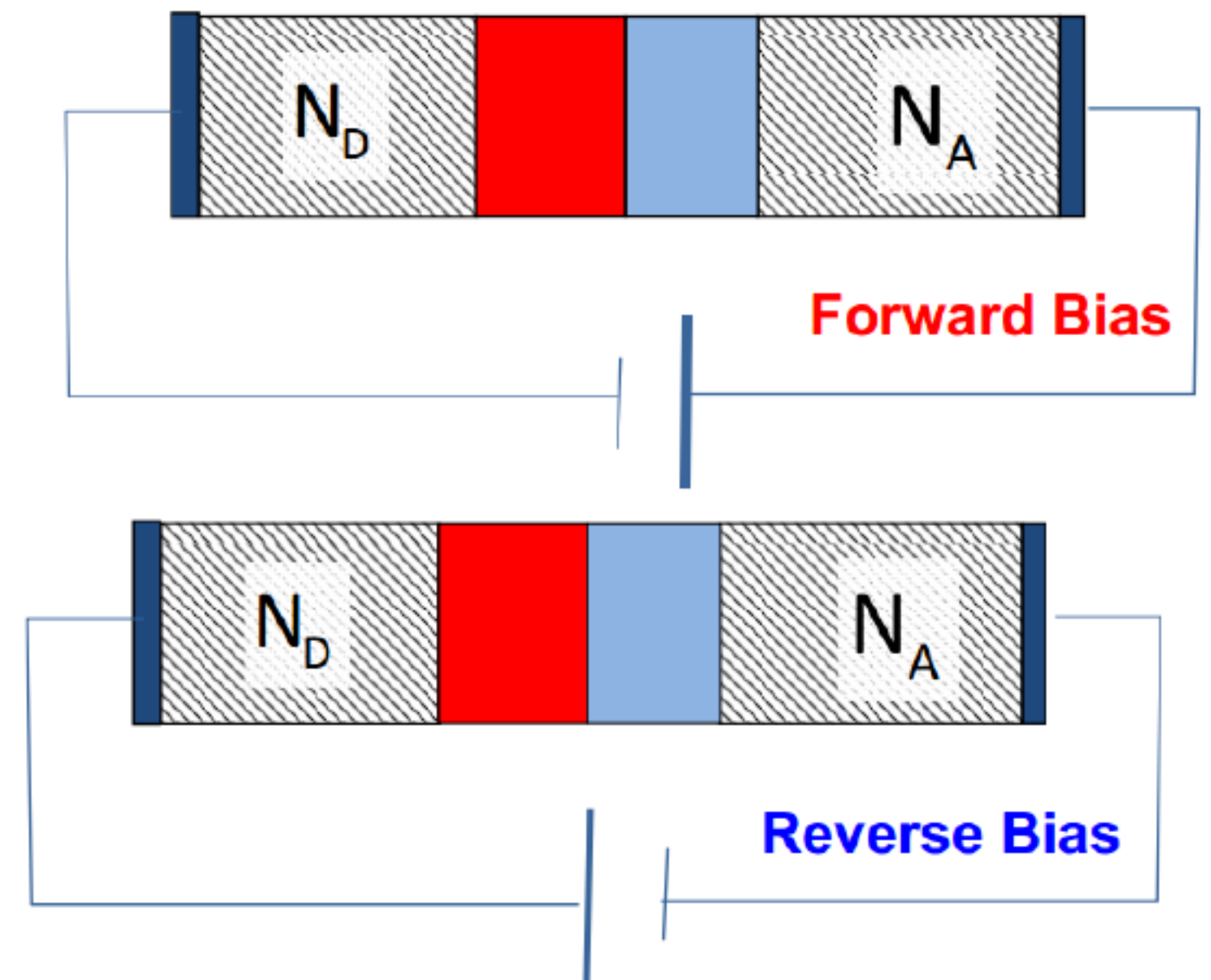
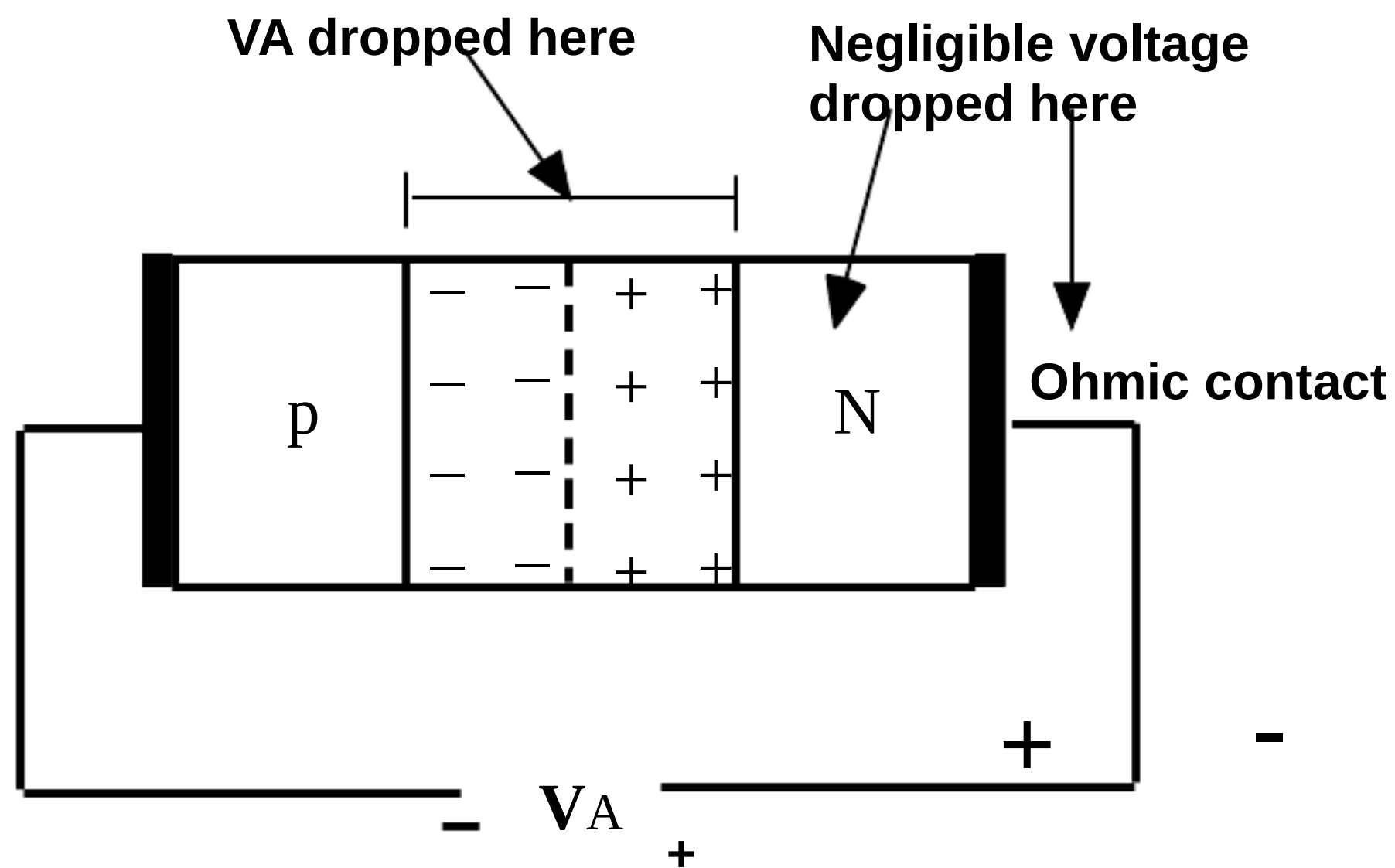
$$x_d \cong x_n = \sqrt{\frac{2\epsilon V_{bi}}{e N_D}}$$

Unbiased junction



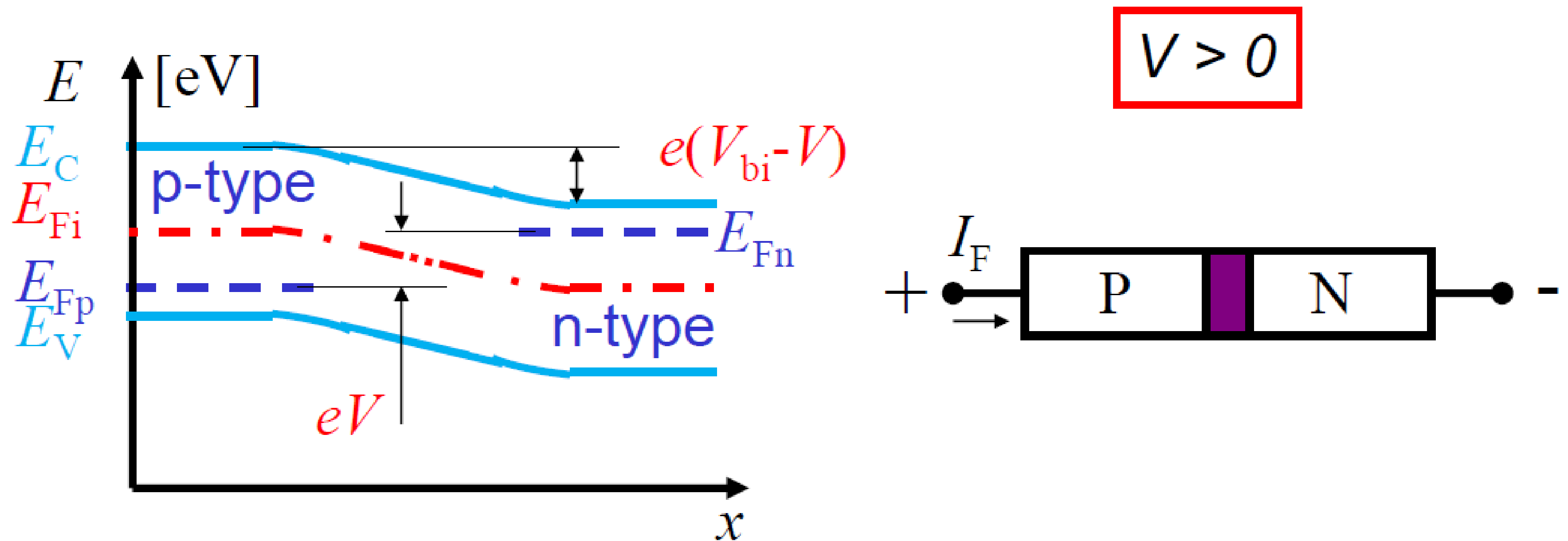
Biased PN Junction

Quiz: where does the voltage drop occur ?



Applied voltage appears essentially entirely across the depletion region. The total potential drop across the junction is , $V_{bi} - V_A$.

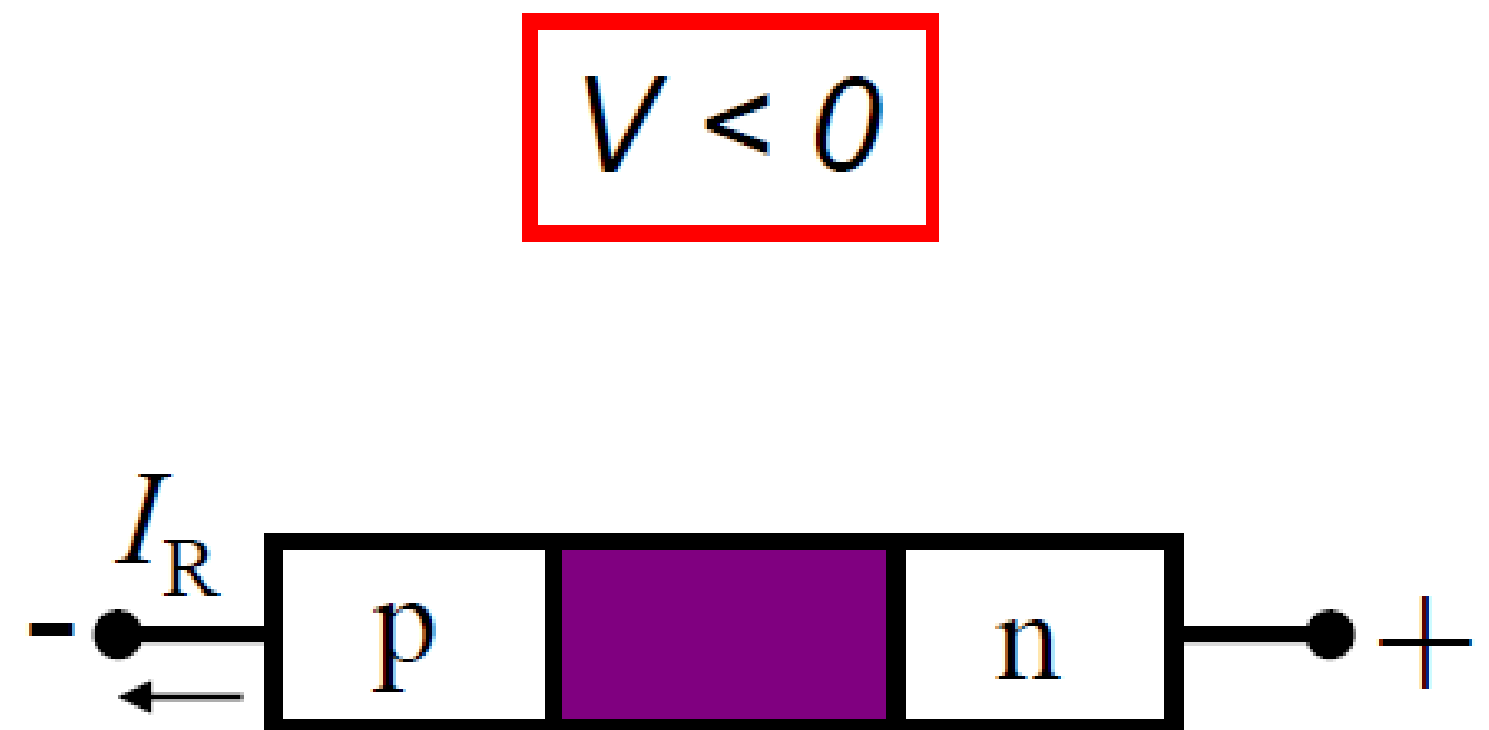
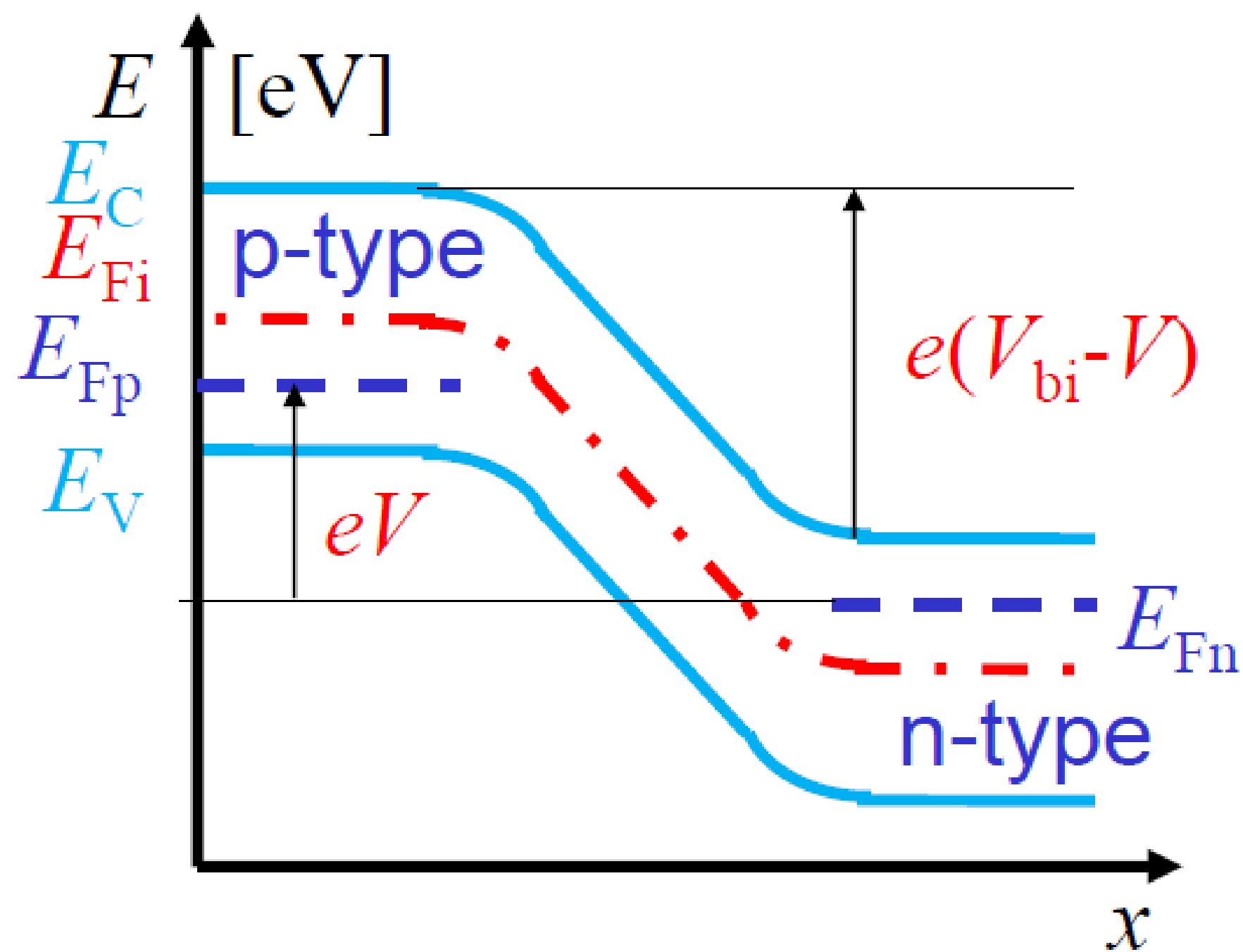
Forward-biased junction



$\mathcal{E}$ -field across depletion region decreases.

W decreases, diffusion pushes mobile carriers towards the junction.

Reverse-biased junction



Depletion layer width for polarized junction

$$x_d = \sqrt{\frac{2\varepsilon}{e} \frac{(N_A + N_D)}{N_D N_A} (V_{bi} - V)}$$

convention: V - is positive for positive on p and negative on n

for one side abrupt junction:

$$x_d = \sqrt{\frac{2\varepsilon(V_{bi} - V)}{eN_B}}$$

N_B – lightly doped bulk concentration

P-N junction under bias

Built-in potential in the depletion region

- is lowered under FB ($V = V_f$), it decreases to $(V_o - V_f)$
- Is increased under RB ($V = -V_r$), it increases to $(V_o + V_r)$

Electric field in the depletion region

- under FB, E decreases (external field opposes built-in field)
- under RB, E increases (external field aids built-in field)

Depletion region with W

- under FB, W decreases
- under RB, W increases

P-N junction under bias

Energy band separation

- under FB, $E_{cp} - E_{cn} = q(V_o - V_f)$ – decreases
- under RB, $E_{cp} - E_{cn} = q(V_o + V_r)$ – increases

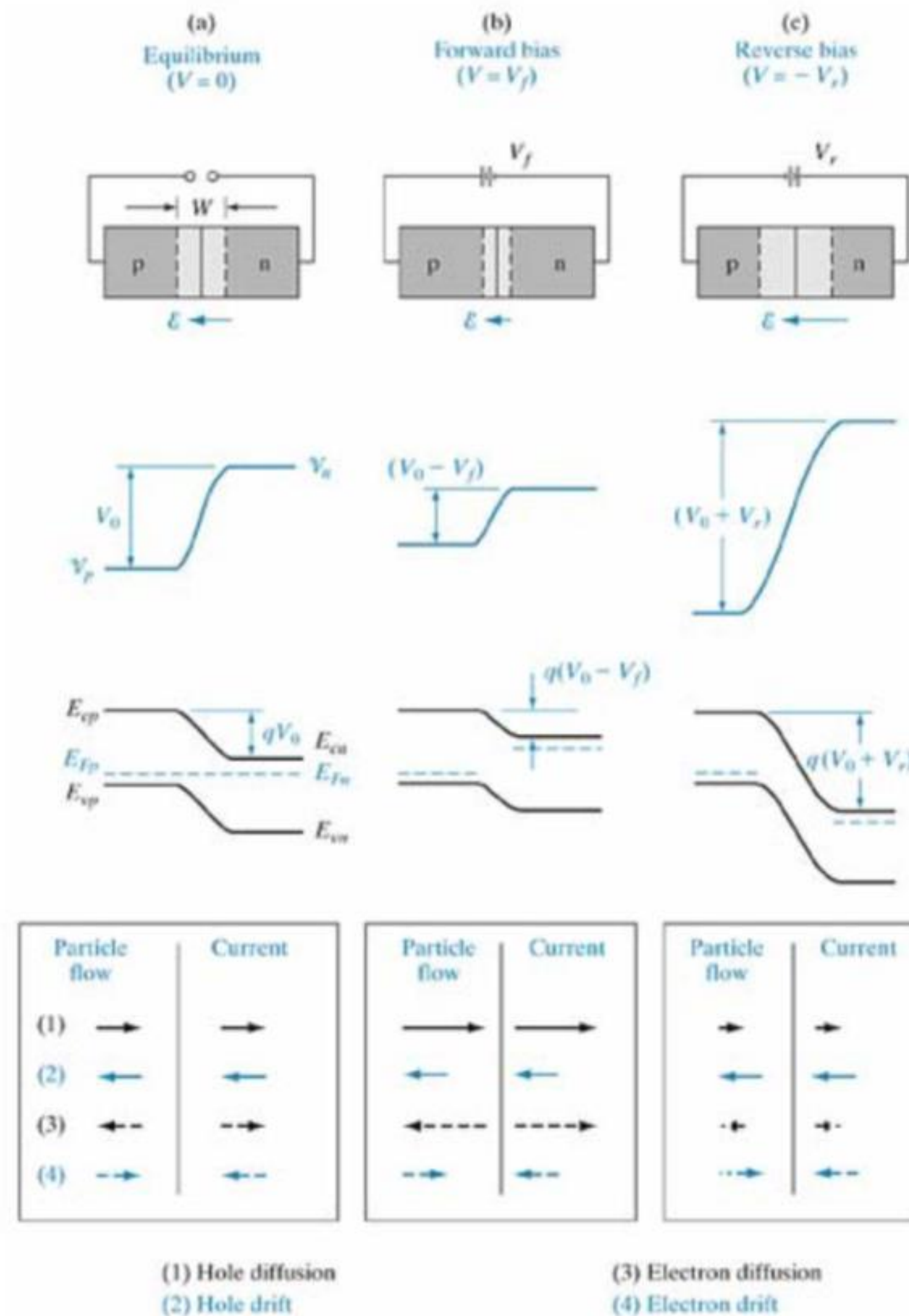
Bands are separated less in FB and more in RB compared to thermal equilibrium condition.

Fermi levels E_{Fp} and E_{Fn} are no longer aligned (applied V results in a non-equilibrium condition). Their separation is numerically equal to the applied voltage, V_f for FB and V_r for RB.

In summary

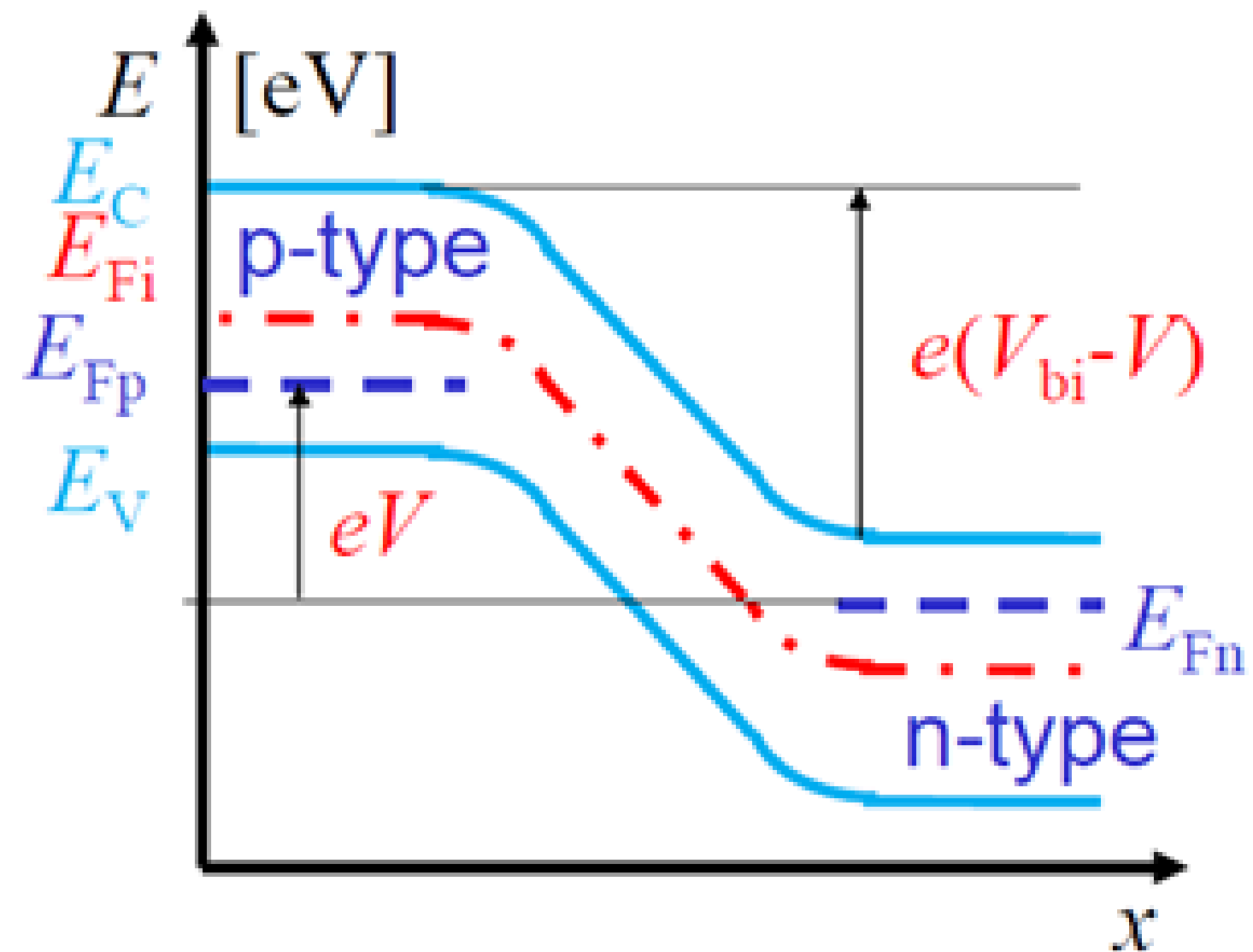
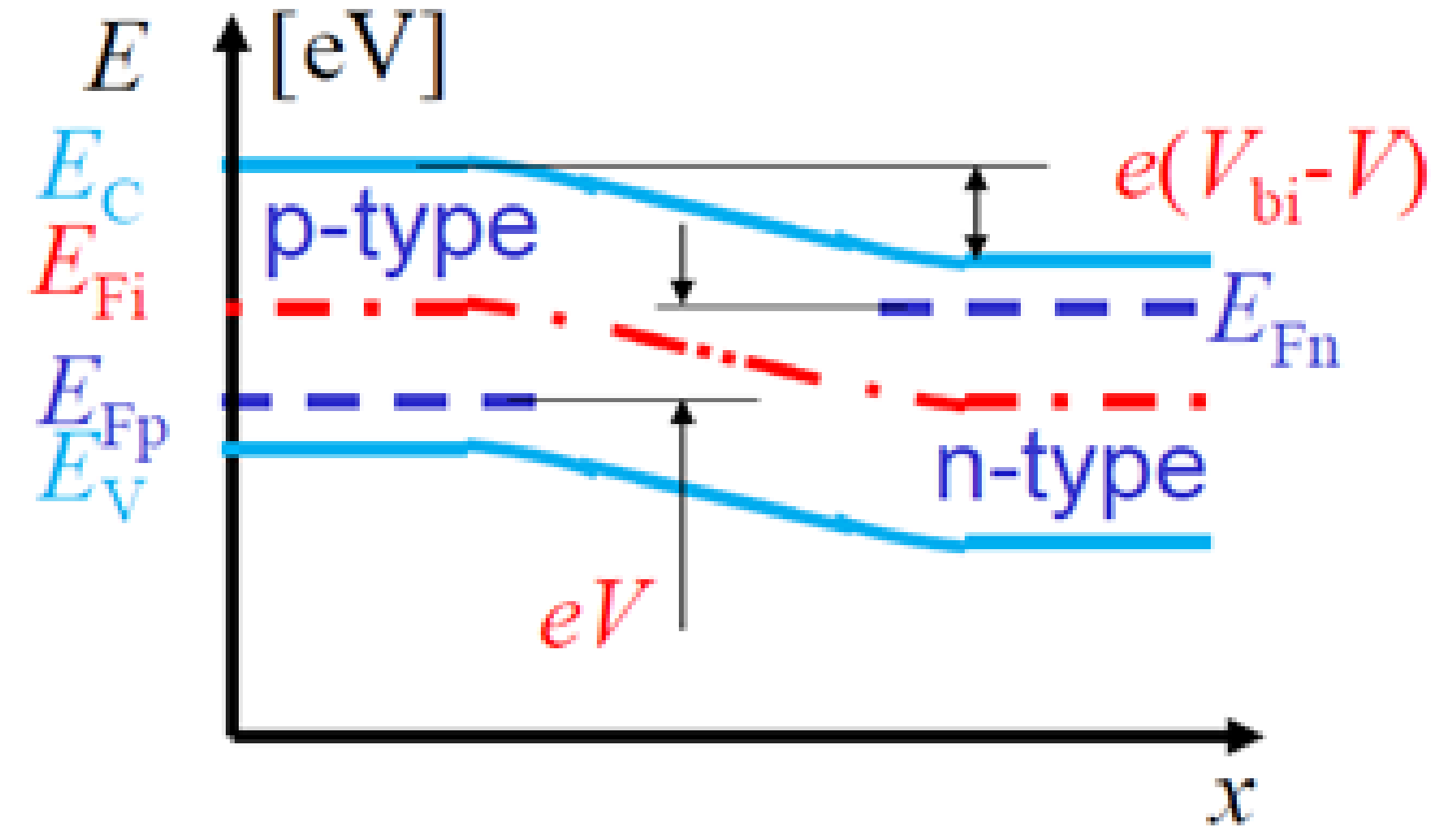
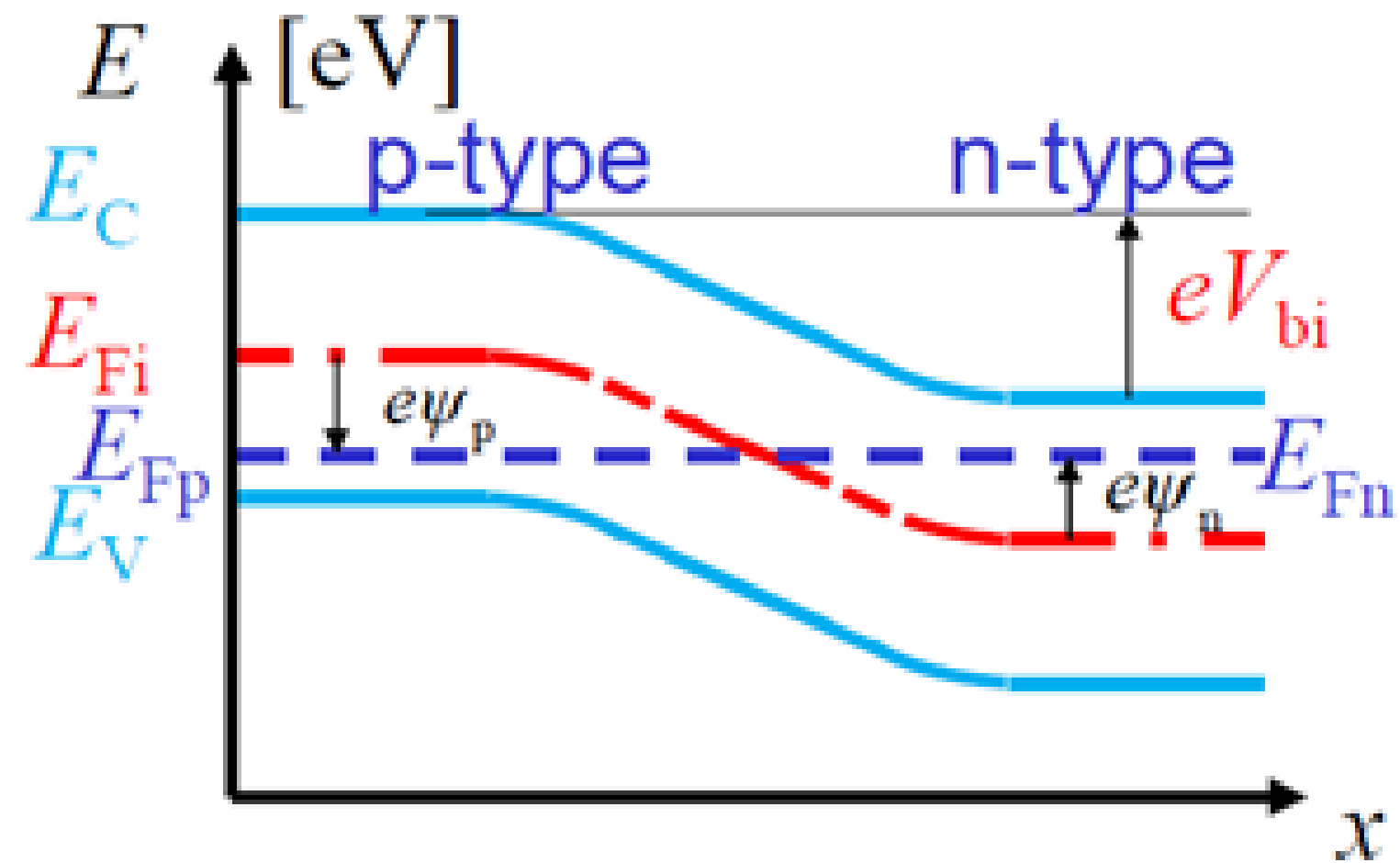
under FB, built-in potential ↓, E ↓, W ↓, energy barrier ↓

under RB, built-in potential ↑, E ↑, W ↑, energy barrier ↑

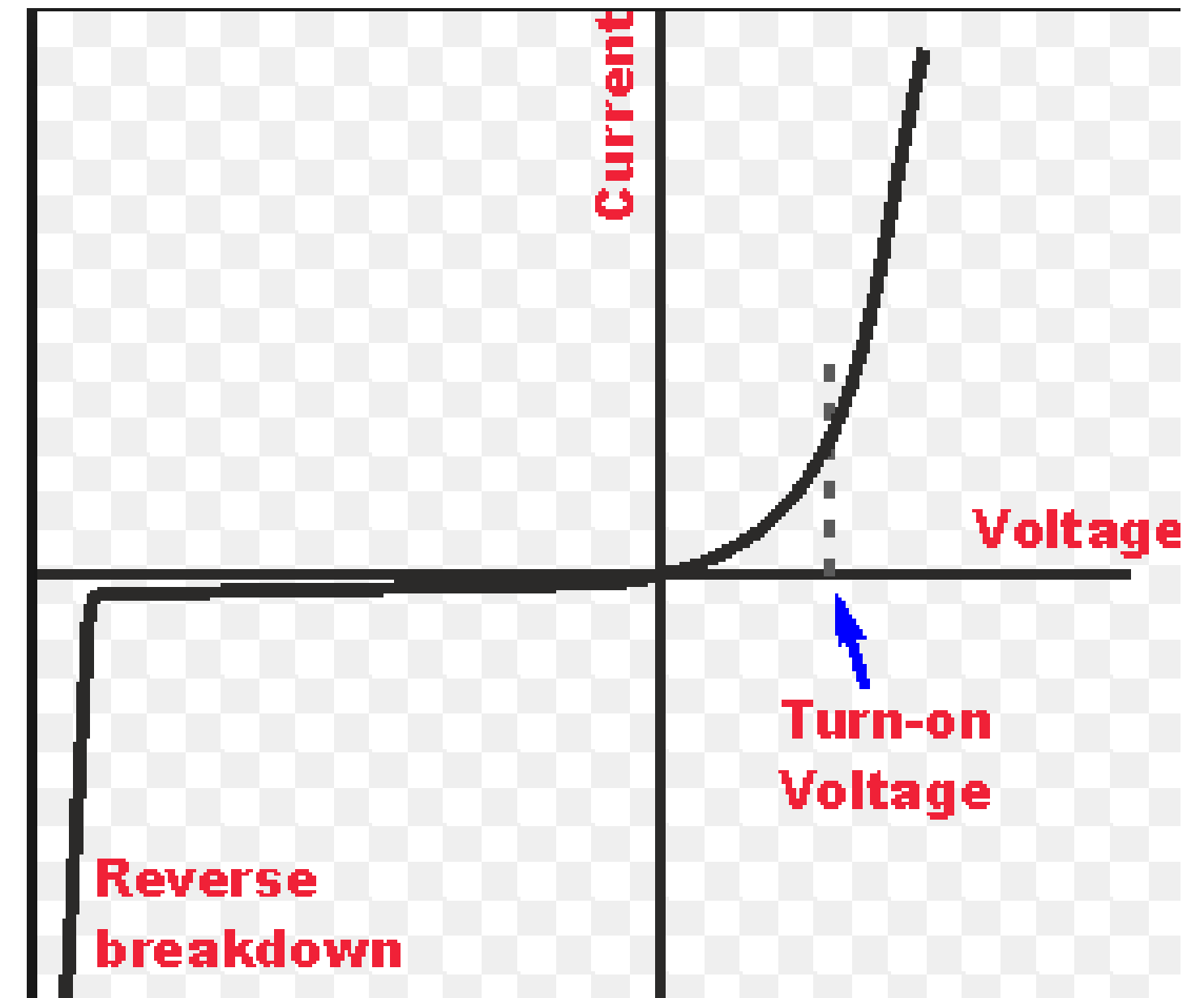
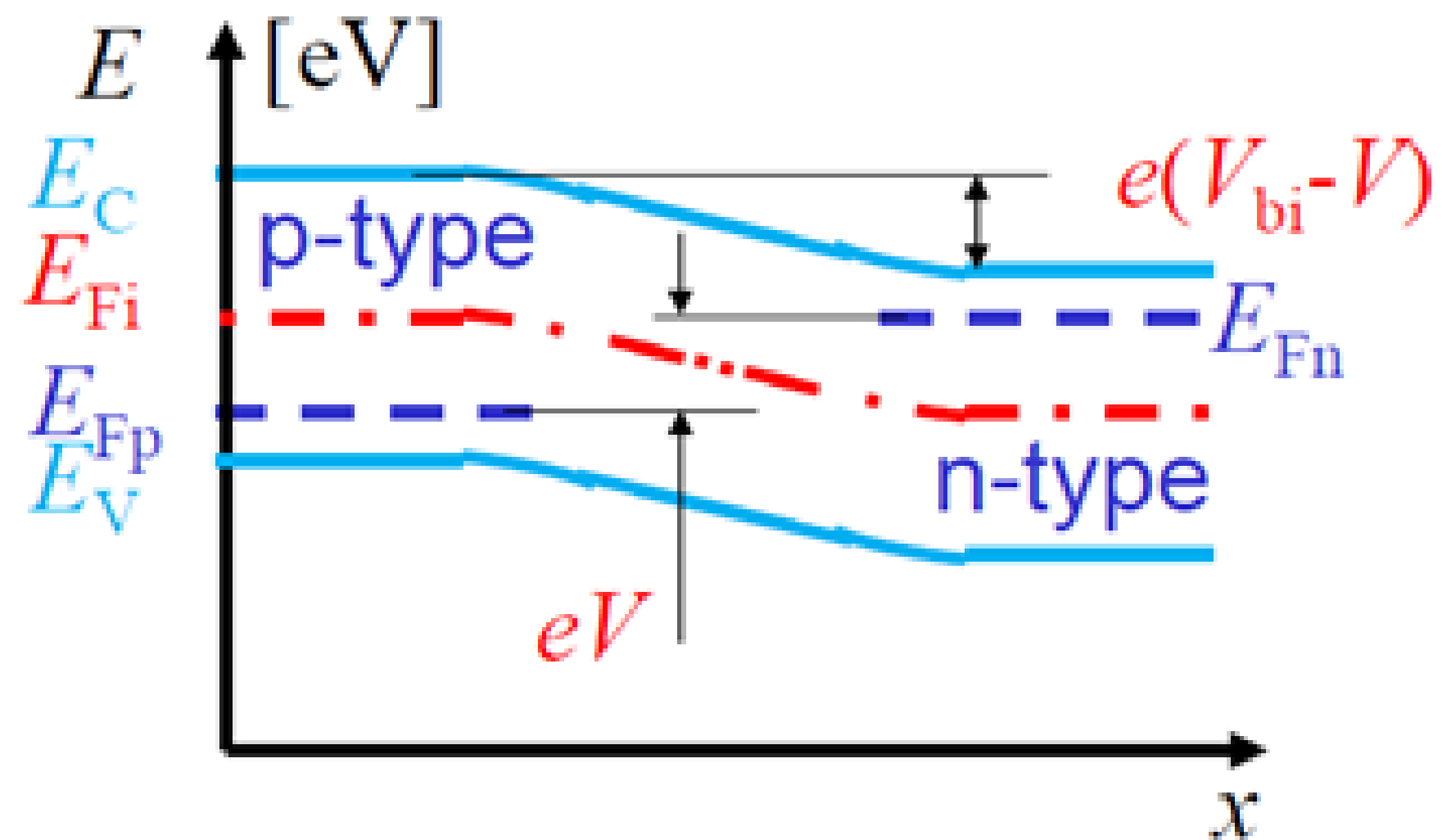


Effects of a bias at a p-n junction; transition region width and electric field, electrostatic potential, energy band diagram, and particle flow and current directions within W for (a) equilibrium, (b) forward bias, and (c) reverse bias.

Quiz: what bias was applied?



Quiz: what happen if external bias is larger than V_{bi} ?



Quiz: width of depletion increase or decrease under:

1. Forward bias
2. Reverse bias
3. Increase N_a or N_d

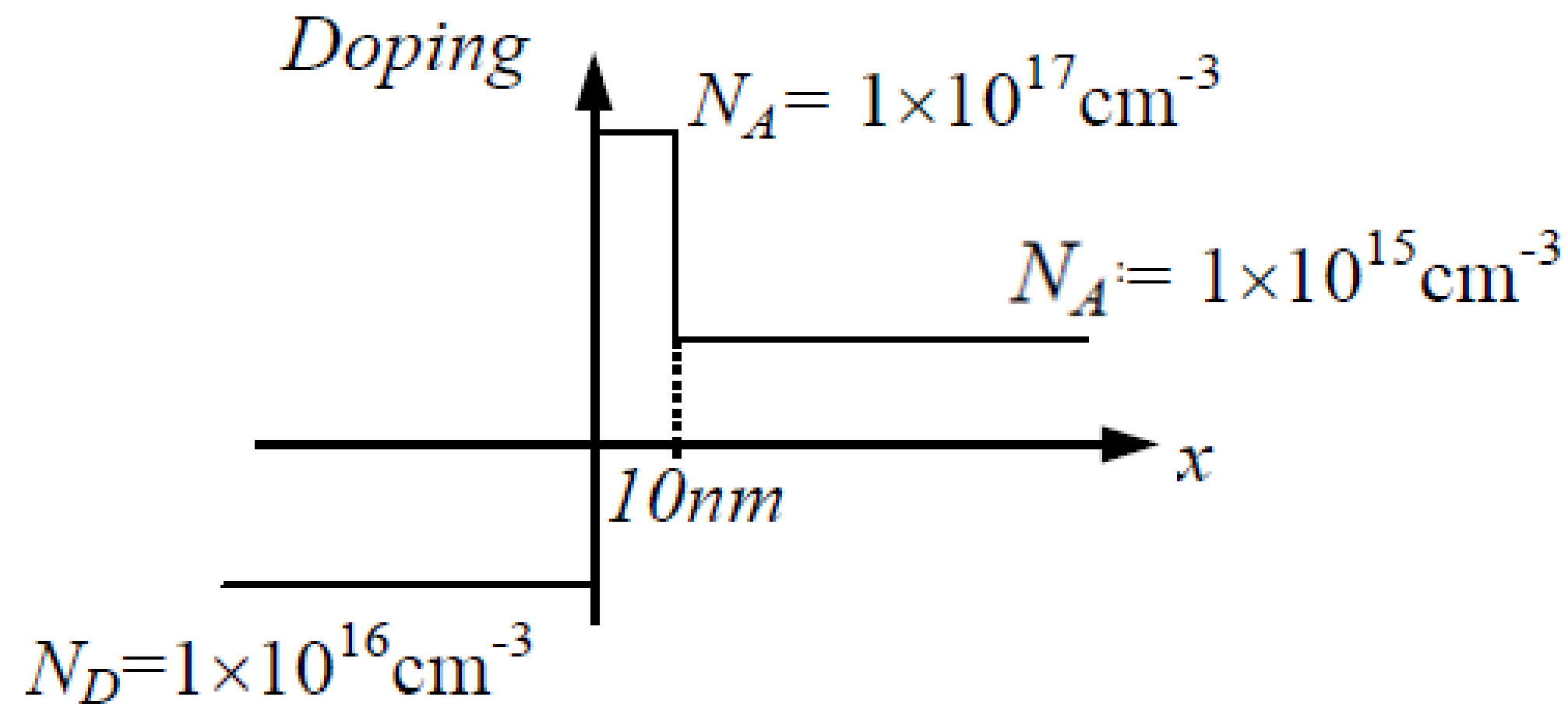
Useful link

http://www.cleanroom.byu.edu/pn_junction.phtml

A p⁺/n silicon step junction is doped with $N_A=10^{19}\text{cm}^{-3}$ and $N_D=5\times 10^{15}\text{cm}^{-3}$.

- a) What is the built-in potential at room temperature?
- b) What is the built-in potential at 800K?
- c) What is the width of the depletion region at the n-side?
- d) What is the total depletion region width of the P-N junction?
- e) What is the maximum electric field in the depletion region?
- f) If a reverse bias of 4V is applied to the diode, what is the new maximum electric field in the junction?
- g) Calculate the percentage increase in the depletion width on the n-side and the p-side under 4V reverse bias.

For a PN Junction with doping profile as shown in the Figure.



Drawing not to scale

- Sketch the E-Field and Potential of the System.
- What is the maximum E-field and built-in potential?